SMART FARMING THROUGH MULTI TEMPORAL CLIMATE BASED AI FOR AGRICULTURAL DECISION SUPPORT

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

BEHARA REVATHI KUSUMA	**22A51A0572**
KONDETI AKHIL LOKESH	**22A51A0596**
KONDAKA TARUN SAI	**23A55A0514**
VALUROUTHU UDAY CHANDRA NAIDU	**22A51A05D0**
KOLLI CHAITHRA JYOTSHNA	**22A51A0594**

Under the Guidance of

Dr. Y. Ramesh **M.Tech , Ph.D.**

HOD, Dept of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ADITYA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(An Autonomous Institution)

Approved by AICTE, Permanently Affiliated to JNTUGV – Vizianagaram

Accredited by NBA & NAAC with A+, Recognized Under 2(f) 12(B) of UGC

K.Kotturu, Tekkali, Srikakulam - 532201, Andhra Pradesh.

April – 2026

DECLARATION

*"We hereby declare that the project entitled **" Smart Farming through Multi Temporal Climate based AI for Agricultural Decision Support"** Submitted for the award of the degree of Bachelor of Technology in* ***Computer Science and Engineering*** *is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the completion for the award of any other degree, associate ship, fellowship or any other similar titles."*

PLACE: Tekkali	**Behara Revathi Kusuma**	**22A51A0572**
DATE :	**Kondeti Akhil Lokesh**	**22A51A0596**
	Kondaka Tarun Sai	**23A55A0514**
	Valurouthu Uday Chandra Naidu	**22A51A05D0**
	Kolli Chaithra Jyotshna	**22A51A0594**

ADITYA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(An Autonomous Institution)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project report entitled "**SMART FARMING THROUGH MULTI TEMPORAL CLIMATE BASED AI FOR AGRICULTURAL DECISION SUPPORT**" being submitted by BEHARA REVATHI KUSUMA (Registered No.:22A51A0572), KONDETI AKHIL LOKESH (Registered No.:22A51A0596), KONDAKA TARUN SAI (Registered No. :23A55A0514), VALUROUTHU UDAY CHANDRA NAIDU (Registered No.:22A51A05D0), KOLLI CHAITHRA JYOTHSNA (Registered No.:22A51A0594). In partial fulfillment for the award of the Degree of Bachelor of Technology in **Computer Science and Engineering** to the Jawaharlal Nehru Technological University Gurajada Vizianagaram is a record of Bonafide work carried out under my guidance and supervision.

Dr. Y. Ramesh M.Tech, Ph.D. **Head of the Department**	**Dr. Y. Ramesh** M.Tech, Ph.D. **Head of the Department**

ACKNOWLEDGEMENT

We wish to thank **Dr. Y**. **Ramesh** for his kind support and his valuable suggestions and encouragement helped us a lot in carrying out this project work as well as in bringing this project to this form.

We take this opportunity to express our sincere gratitude to our Director **Prof. V. V. Nageswara Rao** for his encouragement in all respect.

We take the privilege to thank our principal **Dr. A. S. Srinivasa Rao** for his encouragement and support.

We are also very much thankful to **Dr. Y. Ramesh**, Head of the Department, Computer Science and Engineering for his help and valuable support in completing the project

We are also thankful to all staff members in the Department of Computer Science and Engineering, for their feedback in the reviews and kind help throughout our project

Last but not least, we thank all our classmates for their encouragement and help in making this project a success by maintaining consistency in working out the project during from the data collection phase to implementation phase.

It is their help and support, due to which we were able to complete the design and technical report.

Behara Revathi Kusuma	**22A51A0572**
Kondeti Akhil Lokesh	**22A51A0596**
Kondaka Tarun Sai	**23A55A0514**
Valurouthu Uday Chandra Naidu	**22A51A05D0**
Kolli Chaithra Jyotshna	**22A51A0594**

Project Details

Batch No	**:**	21
Project Title	**:**	Smart Farming through Multi Temporal Climate based AI for Agricultural Decision Support
Area/Domain	**:**	Artificial Intelligence & Machine Learning (Agricultural Data Analytics)
Application	**:**	Smart Farming Decision Support System
Description	**:**	This project develops a smart farming decision support system using Artificial Intelligence and multi-temporal climate data. It analyses important parameters such as rainfall, temperature, and soil moisture obtained from NASA POWER datasets to understand agricultural conditions over time. An Agricultural Condition Level Index (ACLI) is calculated to evaluate land suitability, and machine learning models like Decision Tree and Gradient Boosting are used to classify land into poor, moderate, and good categories. Based on this analysis, the system provides useful recommendations for irrigation planning, crop selection, and seasonal decisions. The approach is cost-effective, scalable, and suitable for rural areas as it relies on freely available climate data instead of expensive sensor-based systems.
Project Outcome	**:**	The proposed system improves agricultural decision-making using multi-temporal climate data and machine learning. The Gradient Boosting model provides higher accuracy than the Decision Tree for reliable land classification into poor, moderate, and good conditions based on ACLI. It offers practical recommendations for irrigation, crop planning, and seasonal decisions. Overall, the system is cost-effective, scalable, and supports sustainable, data-driven farming without requiring expensive hardware.

Batch Details

Project Guide
Dr. Y.Ramesh
HOD,Dept of CSE

Contact No: +91 9849765747
rameshyegireddi@gmail.com

Behara Revathi Kusuma
Regd No: 22A51A0572
revathikusuma65@gmail.com
Contact No: +91 8978182061

Kondeti Akhil Lokesh
Regd No: 22A51A0596
akhilkondeti596@gmail.com
Contact No: +91 6303122196

Kondaka Tarun Sai
Regd No: 23A55A0514
kondakatarunsai@gmail.com
Contact No: +91 9390295920

Valurouthu Uday Chandra Naidu
Regd No: 22A51A05D0
valurouthuudaychandranaidu@gmail.com
Contact No: +91 8074894799

Kolli Chaithra Jyotshna
Regd No: 22A51A0594
chaithra.kolli17@gmail.com
Contact No: +918332927609

Program Outcomes (PO)

1. **ENGINEERING KNOWLEDGE:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **PROBLEM ANALYSIS:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **DESIGN/DEVELOPMENT OF SOLUTIONS:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **CONDUCT INVESTIGATIONS OF COMPLEX PROBLEMS:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **MODERN TOOL USAGE:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **THE ENGINEER AND SOCIETY:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **ENVIRONMENT AND SUSTAINABILITY:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **ETHICS:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **INDIVIDUAL AND TEAM WORK:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **COMMUNICATION:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **PROJECT MANAGEMENT AND FINANCE:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **LIFE-LONG LEARNING:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

- **PSO1:-** Apply mathematical foundations, algorithmic principles, and theoretical computer science in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices.
- **PSO2:-** Demonstrate understanding of the principles and working of the hardware and software aspects of computer systems.
- **PSO3:-** Use knowledge in various domains to identify research gaps and hence to provide solution to new ideas and innovations

PO-PSO Mapping

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
3	3	3	3	3	2	2	1	3	2	3	3	3	3	2

ABSTRACT

Agricultural production is largely dependent upon the climate, but the fact is many farmers cannot afford the cost of soil analysis equipment, which is necessary to check the nutrient level. However, as a solution to the above problem, this project would outline the procedure of determining the level of agricultural land by employing climate variables as opposed to the chemical composition of the soil. Here, data regarding the level of moisture present in the root zone of the soil, the wetness of the soil, rainfall, and the temperature of the land are calculated to obtain a value referred to as the Agriculture Condition Level Index (ACLI). Rather than depending on the actual production of crops, the agricultural land is classified into three categories, namely poor, moderately, and good, depending upon the pre-defined ACLI value. Decision Tree is used as the baseline model for capturing the interaction between the environment variables and crop growth, while Gradient Boosting is used for better predictions through the use of ensemble learning. It can be seen that, in general, the land condition during the monsoon season is quite good due to sufficient rainfall, while land conditions in the summer season reveal higher climate stress. This approach is relatively cost-effective and practical since it does not require chemical soil nutrient data such as NPK, hence appropriately applicable to resource-constrained farmers. Future works include the incorporation of aerial images and market information into the system for a multimodal precision farming system.

Keywords-- Precision Agriculture, Climate based Decision Support, Multi temporal climate analysis, Machine Learning in Agriculture, Crop stress prediction, Irrigation Planning.

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep learning
RNN	Recurrent neural networks
ACLI	Agricultural Condition Level Index
RZSW	Root Zone Soil Wetness
SSW	Surface Soil Wetness
RMSE	Root Mean Square Error
CNN	Convolutional Neural Network
MAE	Mean Absolute Error
R^2	coefficient of determination
GPU	Graphical Processing Unit
CPU	Central Processing Unit
RAM	Random Access Memory
HDD	Hard Disk Drive
SSD	Solid State Drive
GB	Gigabyte
GTX	Giga Texel Shader eXtreme
CUDA	Compute Unified Device Architecture
OS	Operating System
RH	Relative Humidity
SM	Soil Mositure
DT	Decision Tree
RF	Random Forest
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory
XGBoost	Extreme Gradient Boosting
IoT	Internet of Things

NASA	National Aeronautics and Space Administration
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R^2	Coefficient of Determination
PO	Program Outcomes
PSO	Program Specific Outcomes
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
MBE	Mean Bias Error
CE	Coefficient of Efficiency
CC	Correlation Coefficient
RB	Relative Bias
R	Correlation Coefficient

LIST OF SYMBOLS

Symbol	Meaning
R	Rainfall
T	Temperature
RH	Relative Humidity
SM	Soil Moisture
RZSW	Root Zone Soil Wetness
SSW	Surface Soil Wetness
ACLI	Agricultural Condition Level Index
I_i (I)	Irrigation Index
C_i (C)	Crop Index
n	Number of Samples
μ (mu)	Mean
σ (sigma)	Standard Deviation
Σ (sigma)	Summation
√	Square Root
$(y_i - \hat{y}_i)$	Error term
$(y_i - \hat{y}_i)^2$	Squared Error
**	$y_i - \hat{y}_i$
R^2	Coefficient of Determination
P	Precision
R (metric)	Recall
F1	F1 Score
H(S)	Entropy
IG	Information Gain
p_i	Probability of class i
$\log_2$	Logarithm base 2
$f_m(x)$	Weak learner at stage m
$F_m(x)$	Final model at stage m
η (eta)	Learning Rate
m	Iteration number
DOY	Day of Year
Date	Time index
min(x)	Minimum Value
max(x)	Maximum Value

CHAPTER – 1

INTRODUCTION

1.1 Introduction to Smart Farming and Climate-Based Decision Support

Agriculture plays a crucial role in the economic and social development of many countries, especially in developing nations like India, where a significant portion of the population depends on farming for their livelihood. It is one of the primary sources of food, employment, and income, contributing significantly to the national economy. The productivity and sustainability of agriculture are highly influenced by environmental and climatic factors such as temperature, rainfall, soil moisture, humidity, and seasonal variations. These factors directly affect crop growth, soil fertility, water availability, and overall agricultural output.

In recent years, climate change has introduced significant challenges in the agricultural sector. Irregular rainfall patterns, rising temperatures, extreme weather events such as droughts and floods, and seasonal uncertainties have made traditional farming practices less reliable. Farmers often face difficulties in making timely and informed decisions regarding irrigation scheduling, crop selection, fertilizer usage, and harvesting periods. These uncertainties can lead to reduced crop yield, financial losses, and inefficient use of natural resources.

Traditional agricultural practices largely rely on farmers' experience, intuition, and historical knowledge. While these methods have been effective in the past, they are no longer sufficient to handle the increasing complexity and variability of modern climatic conditions. Moreover, access to advanced technologies such as soil testing laboratories and IoT-based monitoring systems is limited in many rural areas due to high costs and lack of technical expertise.

To overcome these challenges, the concept of smart farming has gained significant importance. Smart farming involves the use of modern technologies such as Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and data analytics to enhance agricultural productivity and efficiency. These technologies enable the collection, processing, and analysis of large volumes of environmental and agricultural data to generate meaningful insights.

Through smart farming, data-driven approaches replace traditional intuition-based methods, allowing farmers to make more accurate and timely decisions. Machine learning models can analyze patterns in climate data, predict future conditions, and provide recommendations for optimal farming practices. This leads to improved crop yield, better resource management, and reduced risks associated with climate variability.

A key component of smart farming is the development of climate-based decision support systems. These systems utilize environmental data such as rainfall, temperature, soil moisture, and humidity to evaluate agricultural conditions and assess land suitability. By integrating multi-temporal climate data with machine learning algorithms, these systems can classify agricultural land, predict crop stress, and provide recommendations for irrigation planning and crop selection.

Furthermore, climate-based decision support systems help bridge the gap between data analysis and practical agricultural decision-making. Instead of only providing predictions, these systems offer actionable insights that farmers can directly apply in their daily activities. This improves decision accuracy, reduces uncertainty, and enhances overall farm productivity.

Another important advantage of smart farming systems is their scalability and cost-effectiveness. By relying on freely available climate datasets instead of expensive sensor-based infrastructure, these systems can be easily adopted in rural and resource-constrained regions. This makes advanced agricultural technologies more accessible to small and marginal farmers.

In addition, smart farming contributes to sustainable agriculture by promoting efficient use of water, soil, and other natural resources. It supports environmentally friendly practices and helps reduce the negative impact of climate change on agriculture. By enabling data-driven decision-making, smart farming ensures long-term agricultural sustainability and food security.

In conclusion, smart farming combined with climate-based decision support systems provides a powerful solution to the challenges faced by modern agriculture. By leveraging advanced technologies and multi-temporal climate data, it enhances productivity, improves resource utilization, and supports sustainable farming practices. This approach plays a vital role in transforming traditional agriculture into a more efficient, intelligent, and resilient system.

1.2 Challenges in Traditional and Modern Agricultural Practices

Despite the availability of advanced agricultural technologies, several challenges still exist in both traditional and modern farming practices. Traditional agriculture primarily relies on farmers' experience, observation, and historical knowledge. While this approach has been effective in the past, it is no longer sufficient to address the complexities introduced by climate variability, environmental degradation, and changing weather patterns.

One of the major limitations of traditional farming is its dependence on intuition rather than data-driven insights. Farmers often make decisions based on past experiences, which may not be reliable under current climatic conditions. Unpredictable rainfall, temperature fluctuations, and extreme weather events make it difficult to follow fixed farming schedules, leading to reduced productivity and increased risk of crop failure.

Another commonly used scientific approach in agriculture is soil testing, which helps determine the nutrient content of the soil, including nitrogen, phosphorus, and potassium levels. Although soil testing provides valuable insights for improving crop yield, it requires laboratory facilities, specialized equipment, and trained personnel. This makes it expensive and less accessible, especially for small and marginal farmers in rural areas. Additionally, soil testing is not performed frequently due to cost and time constraints, which limits its effectiveness in dynamic agricultural environments.

Modern agricultural systems have introduced advanced technologies such as IoT-based sensors to monitor environmental conditions like soil moisture, temperature, and humidity in real time. These systems provide accurate and continuous data, enabling better decision-making. However, they come with several challenges, including high installation costs, maintenance expenses, and dependency on reliable power supply and internet connectivity. Moreover, farmers need technical knowledge to operate and maintain these systems, which limits their adoption in resource-constrained regions.

Another challenge in modern agriculture is the availability and quality of data. Many advanced systems rely on large volumes of high-quality data for training machine learning models. In many rural areas, such data is either unavailable, incomplete, or inconsistent. This affects the accuracy and reliability of predictive models and reduces their practical applicability.

Furthermore, many existing machine learning models in agriculture focus primarily on prediction tasks such as crop yield estimation, weather forecasting, or stress detection. Although these models achieve high accuracy, they often fail to provide actionable recommendations that farmers can directly use. For example, a model may predict low yield but may not suggest specific actions such as irrigation scheduling, crop selection, or fertilizer application. This creates a gap between data analysis and practical decision-making.

Another important limitation is the lack of long-term climate analysis in many existing systems. Most models focus on short-term predictions and fail to capture seasonal trends and long-term climatic patterns. This limits their ability to support sustainable agricultural planning and adaptation to climate change.

In addition, the complexity of advanced agricultural technologies can be a barrier to adoption. Many systems involve complicated interfaces, data processing techniques, and model interpretations, making them difficult for farmers to understand and use effectively. Lack of training and awareness further reduces their usability in real-world scenarios.
Economic constraints also play a significant role in limiting the adoption of modern agricultural technologies. Small-scale farmers often cannot afford expensive equipment, software tools, or subscription-based services. This creates a digital divide where only large-scale or commercial farmers can benefit from advanced technologies.

Moreover, existing systems often operate in isolation and do not integrate multiple data sources such as climate data, soil data, and market information. This limits their ability to provide comprehensive decision support. A holistic approach that combines different data sources is necessary for better agricultural planning and management.

Overall, the major challenges in traditional and modern agricultural practices include:

- Dependence on experience rather than data-driven decision-making
- High cost and limited accessibility of soil testing
- Expensive and complex IoT-based systems
- Lack of reliable and high-quality data
- Focus on prediction rather than actionable decision support
- Limited consideration of long-term climatic patterns
- Complexity and lack of user-friendly systems
- Economic constraints for small and marginal farmers
- Lack of integration across different agricultural data sources

These challenges highlight the need for a cost-effective, scalable, and intelligent solution that can utilize readily available data and provide practical decision support. Addressing these limitations is essential for improving agricultural productivity, ensuring sustainability, and supporting farmers in adapting to changing climatic conditions.

1.3 Review of Existing Approaches and Their Limitations

In recent years, significant research has been carried out in the field of smart agriculture using Artificial Intelligence (AI) and Machine Learning (ML) techniques. Various approaches have been proposed for improving agricultural productivity, including crop yield prediction, crop recommendation systems, drought stress detection, and climate-based analysis. These approaches aim to assist farmers by analyzing environmental data and generating useful insights.

One of the most widely used approaches in agriculture is crop yield prediction using machine learning models such as Random Forest, Support Vector Machine (SVM), and K-Nearest

Neighbors (KNN). These models analyze historical climate and soil data to estimate crop production. While they provide good prediction accuracy, they are primarily focused on forecasting outcomes rather than offering practical decision support. As a result, farmers may not receive clear guidance on what actions to take based on the predictions.

Another important approach involves the use of deep learning models such as Long Short-Term Memory (LSTM) networks for time-series forecasting of climate variables. These models are effective in capturing temporal patterns and short-term dependencies in weather data. However, many of these models are trained on limited time windows and fail to capture long-term seasonal variations and climate trends, which are essential for sustainable agricultural planning.

Satellite-based and remote sensing approaches have also been widely explored for monitoring agricultural conditions. These methods use data from satellites such as Sentinel and other remote sensing platforms to assess vegetation health, soil moisture, and crop conditions over large areas. Although these approaches provide large-scale insights, they often require complex preprocessing, specialized tools, and high computational resources. This makes them difficult to implement in rural or resource-limited environments.

Some studies have proposed hybrid models that combine multiple machine learning or deep learning techniques to improve prediction accuracy. For example, combinations of Random Forest with LSTM or Gradient Boosting models have shown improved performance in certain applications. However, these hybrid models increase system complexity and require extensive parameter tuning, making them less practical for real-world deployment.

IoT-based smart agriculture systems represent another category of existing approaches. These systems use sensors to collect real-time data on soil moisture, temperature, and humidity. While they provide accurate and real-time monitoring, they involve high installation and maintenance costs. Additionally, they depend on reliable internet connectivity and power supply, which may not be available in many rural areas.

Another limitation observed in existing approaches is their strong dependence on specific types of data, such as soil nutrient data or sensor-based measurements. Many systems require chemical soil analysis (e.g., NPK values) or continuous sensor inputs, which increases cost and reduces accessibility. This limits the adoption of such systems among small and marginal farmers.

Furthermore, most existing systems focus on single-task solutions, such as yield prediction, stress detection, or weather forecasting. They do not provide a comprehensive decision support system that integrates multiple aspects of agriculture. As a result, farmers still need to interpret the results and make decisions on their own, which may lead to inefficiencies.

Another critical limitation is the lack of multi-temporal climate analysis in many studies. Most approaches rely on short-term data and do not consider long-term historical trends or seasonal variations. This reduces the reliability of predictions and limits the ability to support long-term agricultural planning.

In addition, many existing systems lack user-friendly interfaces and practical applicability. The outputs generated by machine learning models are often technical and difficult for farmers to interpret. Without clear and actionable recommendations, the usefulness of these systems in real-world farming remains limited.
From the above analysis, the major limitations of existing approaches can be summarized as follows:

- Focus on prediction rather than actionable decision support
- Limited use of long-term and multi-temporal climate data
- Dependence on expensive infrastructure such as IoT sensors and soil testing
- High computational complexity in deep learning and hybrid models
- Lack of accessibility for small and marginal farmers
- Limited integration of multiple agricultural factors
- Difficulty in interpreting model outputs for practical use

These limitations highlight the need for a more practical, cost-effective, and scalable solution. The proposed system addresses these challenges by utilizing multi-temporal climate data and machine learning models to develop a smart farming decision support system. Unlike existing approaches, it focuses not only on prediction but also on providing meaningful recommendations for irrigation planning, crop selection, and seasonal decision-making.

1.4 Motivation and Need for Multi-Temporal Climate Analysis

Agricultural systems are highly dynamic and are continuously influenced by temporal variations in climatic conditions. Factors such as rainfall, temperature, soil moisture, and humidity vary significantly over time and directly impact crop growth, soil fertility, and agricultural productivity. A single snapshot of environmental data is not sufficient to capture these variations or to understand the overall behavior of agricultural systems. Therefore, it becomes essential to analyze climate data across multiple time scales, including daily, monthly, and seasonal levels.

Multi-temporal climate analysis plays a crucial role in understanding both short-term fluctuations and long-term trends in environmental parameters. Short-term variations help in identifying immediate changes such as sudden rainfall or temperature spikes, while long-term trends provide insights into seasonal patterns, drought cycles, and climate change effects. By combining these perspectives, multi-temporal analysis enables a more accurate and comprehensive assessment of agricultural conditions.

Traditional agricultural practices often rely on farmers' experience and intuition, which may not be sufficient in the presence of unpredictable climatic variations. Moreover, conventional methods such as soil testing and IoT-based monitoring systems can be expensive and difficult to implement in rural or resource-constrained environments. This creates a need for alternative approaches that are cost-effective, scalable, and easy to deploy.

Freely available climate datasets, such as those provided by NASA POWER, offer valuable historical and real-time environmental data over extended periods. These datasets include important parameters such as rainfall, temperature, and soil moisture, which are essential for agricultural analysis. By utilizing such publicly available data, it is possible to eliminate the dependency on expensive hardware and sensor-based systems.

The integration of multi-temporal climate data with machine learning techniques enables the development of intelligent decision support systems. Machine learning models can analyze complex relationships between climatic variables and agricultural conditions, identify hidden patterns, and generate accurate predictions. This helps in classifying land suitability, predicting crop stress, and providing recommendations for irrigation planning and crop selection.

Another important motivation for this study is to bridge the gap between data analysis and practical decision-making in agriculture. Many existing systems focus only on prediction without providing actionable insights. In contrast, the proposed approach aims to deliver meaningful recommendations that can directly assist farmers in improving productivity and resource management.
Furthermore, the use of multi-temporal analysis ensures that the system is robust and adaptable to changing climatic conditions. It enhances the reliability of predictions and supports sustainable farming practices by promoting efficient use of water and other resources.

In summary, the motivation behind this project is to develop a smart farming decision support system that leverages multi-temporal climate data and machine learning techniques to provide accurate, cost-effective, and accessible solutions for agricultural decision-making. This approach aims to improve productivity, reduce risks associated with climate variability, and support sustainable agriculture, especially in rural and resource-limited regions.

1.5 Overview of the Proposed Smart Farming System

The proposed system is a Smart Farming Decision Support System based on Multi-Temporal Climate Analysis, designed to assist farmers in making informed agricultural decisions using data-driven techniques. The system utilizes key climate parameters such as Root Zone Soil

Wetness (RZSW), Surface Soil Wetness (SSW), rainfall, and temperature to evaluate the condition of agricultural land over time.

Unlike traditional approaches that rely on single-time observations, the proposed system analyzes multi-temporal climate data to capture both short-term variations and long-term trends in environmental conditions. This enables a more accurate and reliable assessment of agricultural suitability, helping farmers adapt to changing climatic conditions.

A key contribution of this system is the introduction of the Agricultural Condition Level Index (ACLI). This index integrates multiple climate parameters into a single representative value that reflects the overall condition of the agricultural land. By combining different environmental factors, ACLI simplifies complex data into an easily interpretable metric, making it useful for practical decision-making.

Based on the computed ACLI value, the agricultural land is classified into three categories:

- Poor (low suitability) – Indicates unfavorable conditions for crop growth
- Moderate (medium suitability) – Indicates average conditions with manageable risks
- Good (high suitability) – Indicates optimal conditions for cultivation

This classification helps farmers quickly understand land conditions and make appropriate decisions regarding crop selection, irrigation planning, and resource allocation.
The system employs machine learning techniques to perform classification and prediction tasks. A Decision Tree model is used as a baseline model due to its simplicity, interpretability, and ability to generate clear decision rules. Decision Trees are easy to understand and provide transparent insights into how different climate parameters influence agricultural conditions.
To enhance prediction accuracy and model performance, a Gradient Boosting model is implemented. Gradient Boosting is an ensemble learning technique that combines multiple decision trees to form a strong predictive model. By iteratively correcting errors of previous models, it improves accuracy, reduces bias, and handles complex relationships between variables effectively.

In addition to model development, the system includes data preprocessing steps such as normalization, feature engineering, and handling of missing values to ensure data quality and consistency. Multi-temporal data is structured in a way that preserves temporal dependencies, allowing the models to learn patterns effectively.
Another important feature of the proposed system is its cost-effectiveness and scalability. Instead of relying on expensive IoT sensors or laboratory-based soil testing, the system uses freely available climate datasets, making it accessible to farmers in rural and resource-constrained regions.

Furthermore, the system is designed to provide actionable insights rather than just predictions. By interpreting ACLI values and classification results, farmers can make informed decisions related to irrigation scheduling, crop planning, and land management. This bridges the gap between data analysis and real-world agricultural practices.

The overall workflow of the system includes:

- Collection of multi-temporal climate data
- Data preprocessing and feature engineering
- Calculation of ACLI
- Classification using Decision Tree and Gradient Boosting models
- Evaluation of model performance
- Generation of decision support outputs
-

In summary, the proposed smart farming system integrates multi-temporal climate analysis with machine learning techniques to create a reliable and efficient decision support tool. It addresses the limitations of existing approaches by providing a simple, interpretable, and scalable solution that enhances agricultural productivity and supports sustainable farming practices.

1.6 How the Proposed System Addresses Existing Limitations

The proposed smart farming system effectively addresses several limitations associated with both traditional and modern agricultural approaches. Unlike conventional soil testing methods, which require chemical analysis, laboratory facilities, and trained personnel, the proposed system relies entirely on climate-based parameters such as rainfall, temperature, and soil wetness. These parameters are readily available through publicly accessible datasets, making the system cost-effective and easy to implement.

One of the major challenges in modern agriculture is the dependence on IoT-based sensor systems for real-time monitoring. Although these systems provide accurate data, they involve high installation, maintenance, and operational costs. The proposed system eliminates the need for such expensive infrastructure by utilizing freely available multi-temporal climate data. This significantly reduces the financial burden on farmers and makes the solution accessible to small and marginal farmers, especially in rural and resource-constrained regions.

Another important limitation of existing systems is their reliance on short-term data, which fails to capture seasonal trends and long-term climatic patterns. The proposed system overcomes this issue by incorporating multi-temporal climate analysis, which considers data across different time scales. This allows the system to identify both short-term fluctuations and long-term trends, resulting in more reliable and consistent predictions. As a result, farmers can make better decisions regarding crop planning and resource management.

In many existing machine learning-based agricultural systems, the focus is primarily on prediction rather than practical decision support. These systems often provide outputs such as yield estimates or stress detection without suggesting actionable steps. The proposed system addresses this gap by introducing the Agricultural Condition Level Index (ACLI), which simplifies complex climate data into a single interpretable value. Based on ACLI, the system classifies land into categories such as Poor, Moderate, and Good, enabling farmers to easily understand the condition of their land and take appropriate actions.

The integration of machine learning models further enhances the system's capability to analyze complex relationships between multiple climate variables. The use of a Decision Tree model provides clear and interpretable decision rules, while the Gradient Boosting model improves prediction accuracy by combining multiple weak learners. This combination ensures both transparency and high performance, making the system both reliable and user-friendly.

Another key advantage of the proposed system is its simplicity and scalability. Unlike complex hybrid models or deep learning systems that require high computational resources, the proposed approach uses efficient algorithms that can be implemented on standard computing systems. This makes it suitable for deployment in real-world agricultural environments without requiring advanced infrastructure.

The system also addresses the issue of data accessibility. Instead of relying on specialized or proprietary datasets, it utilizes openly available climate data, which ensures wider applicability and ease of adoption. This reduces dependency on external agencies and enables farmers and researchers to use the system without additional cost.

Furthermore, the proposed system bridges the gap between data analysis and real-world agricultural practices by providing meaningful and actionable insights. It not only analyzes environmental conditions but also supports decision-making related to irrigation planning, crop suitability, and land management. This makes the system more practical and beneficial compared to existing approaches that focus only on analysis.

In addition, the system promotes sustainable agriculture by encouraging efficient use of natural resources such as water and soil. By providing accurate and timely recommendations, it helps reduce resource wastage and supports environmentally friendly farming practices.
Overall, the proposed system addresses the key limitations of existing agricultural approaches by being cost-effective, accessible, scalable, and decision-oriented. It leverages multi-temporal climate data and machine learning techniques to provide a reliable and

intelligent solution for modern agriculture, thereby improving productivity and supporting sustainable farming.

1.7 Objectives of the Study

The primary objective of this project is to develop an intelligent and cost-effective smart farming system that supports agricultural decision-making using climate-based data and machine learning techniques. The system aims to assist farmers in understanding environmental conditions more effectively and making informed decisions to improve crop productivity, resource utilization, and sustainability.

The study focuses on utilizing multi-temporal climate data to capture both short-term variations and long-term trends in environmental parameters. By integrating this data with machine learning models, the system is designed to analyze complex relationships between climate variables and agricultural conditions. A key contribution of this work is the development of the Agricultural Condition Level Index (ACLI), which simplifies multiple climate parameters into a single interpretable value. Based on this index, the system classifies land suitability and provides meaningful recommendations for irrigation planning, crop selection, and land management. Overall, the proposed system aims to bridge the gap between data analysis and practical agricultural decision-making while ensuring accessibility and cost-effectiveness.

Specific Objectives:

- To analyze multi-temporal climate data
- To preprocess and structure climate data for analysis
- To compute the Agricultural Condition Level Index (ACLI)
- To classify land suitability into different categories
- To implement Decision Tree and Gradient Boosting models
- To evaluate and compare model performance
- To provide recommendations for irrigation and crop planning
- To develop a cost-effective and scalable solution
- To promote sustainable agricultural practices

1.8 Scope of the Study

This project focuses on the use of historical climate data (2015–2024) obtained from NASA POWER. The system evaluates agricultural land conditions based on climate parameters and provides decision support for farming activities.

The scope of the project does not include real-time sensor data collection or direct soil nutrient analysis. However, the system can be extended in the future by incorporating additional data sources such as satellite imagery and market information.

CHAPTER – 2

LITERATURE SURVEY

This chapter contains the review of various research works related to smart farming, crop yield prediction, climate-based agricultural analysis, and decision support systems. The study of existing methods helps in understanding the current advancements in the field of agriculture using Artificial Intelligence (AI) and Machine Learning (ML) techniques.
Several researchers have proposed different models and approaches for improving agricultural productivity, including crop recommendation systems, stress detection models, and yield prediction techniques. However, each method has certain limitations in terms of cost, accuracy, scalability, and practical implementation.
This chapter provides a detailed analysis of existing works, their methodologies, performance metrics, and limitations, which helps in identifying the research gap and the need for the proposed system.

2.1 Literature Survey

In recent years, the application of Artificial Intelligence (AI) and Machine Learning (ML) in agriculture has gained significant importance. Various research works have been carried out in areas such as crop yield prediction, crop recommendation, drought stress detection, and climate-based agricultural analysis.

Sharma et al. [1] (2023) proposed a comprehensive machine learning framework for crop yield forecasting in semi-arid zones of Rajasthan, India. The study applied three widely used supervised learning algorithms Random Forest (RF), Support Vector Machine (SVM), and a deep learning neural network to a dataset comprising regional climate and soil information. The RF model achieved the highest accuracy of approximately 92.3%, demonstrating the effectiveness of ensemble learning for agricultural prediction. The authors performed feature importance analysis to identify the most influential variables affecting yield, including temperature, rainfall, and irrigation practices. However, the study was limited in scope in that it focused solely on prediction without incorporating any decision support functionality. The system did not provide actionable recommendations for farmers regarding irrigation scheduling, crop selection, or fertilizer application based on the predicted yield values. This limitation restricts the practical utility of the model for end users.

Mahale et al. [2] (2024) developed a crop recommendation and forecasting system for Maharashtra using a hybrid machine learning approach combining Random Forest with Long Short-Term Memory (LSTM) networks. The RF component handled static feature-based prediction while the LSTM component captured temporal dependencies in weather data. The system achieved an overall accuracy of approximately 92% and demonstrated its ability to recommend suitable crops based on seasonal climate conditions. However, the LSTM component was trained on relatively short historical windows, making it less effective at capturing long-term seasonal patterns and inter-annual climate variability. The system also lacked integration with a comprehensive decision support layer that would allow farmers to receive specific guidance on irrigation timing or crop management practices.

Raeva et al. [3] (2025) conducted a yield prediction study in Bulgaria using multitemporal satellite imagery from Sentinel-2 combined with ERA5 reanalysis climate data. The study applied regression tree models to predict crop yield at field level, achieving an R^2 of 0.78. The use of remote sensing data offered the advantage of spatial coverage, enabling field-level analysis without requiring ground-based measurements. However, the dependency on satellite imagery processing introduces significant technical barriers for widespread adoption. Not all agricultural regions have access to high-quality satellite data processing tools, and the computational resources required for image preprocessing are beyond the reach of most smallholder farmers. The study also did not address the decision support dimension of agricultural management.

Gupta et al. [4] (2025) implemented a crop recommendation system using a combination of RF, SVM, and K-Nearest Neighbors (KNN) algorithms, optimizing the ensemble to achieve approximately 12% accuracy improvement over individual models. The study focused on matching crop recommendations to soil and climate conditions. However, the dataset used was limited in temporal depth, capturing only short-term climate snapshots rather than multi-year historical patterns. The absence of long-term climate analysis limits the system's ability to account for seasonal trends and drought cycles that are critical for reliable agricultural planning.

Ali et al. [5] (2024) addressed the specific problem of drought stress detection in crops using a combination of Gradient Boosting and LSTM models. The system achieved a classification accuracy of approximately 97%, demonstrating high sensitivity to stress conditions derived from satellite-based vegetation indices and meteorological data. While impressive, the narrow application focus on stress detection — without extending the system to support irrigation decisions, crop selection, or other management activities — limits its practical utility as a comprehensive decision support tool. Drought detection as a binary classification problem, while important, represents only one component of the agricultural management lifecycle.

Dhanaraj et al. [6] (2025) proposed an IoT-based on-device AI system for climate-resilient crop yield prediction using lightweight machine learning models optimized for edge deployment. The RF model achieved approximately 90.1% accuracy in yield prediction. The

system's ability to run on low-power IoT devices represents an important contribution toward edge computing in agriculture. However, the fundamental limitation remains the dependency on physical sensor infrastructure. IoT devices require installation, maintenance, calibration, and connectivity, all of which incur costs that are prohibitive for large-scale deployment among resource-constrained farmers.
Mana et al. [7] (2024) presented a comprehensive review of AI and data analytics applications in sustainable agriculture. The study identified key trends in precision farming, including the growing use of ML for crop management, the potential of drone-based monitoring, and the role of big data platforms in agricultural decision-making. However, being a review article, it did not propose or evaluate a specific model, and its findings were largely conceptual rather than empirical. The study highlighted the need for more practically oriented systems that bridge the gap between AI research and on-ground farming realities.

El Hachimi et al. [8] (2023) developed a smart weather data management framework combining LSTM and XGBoost models with ERA5 climate data. The framework was designed to support weather prediction and agricultural decision-making by learning temporal patterns in high-resolution climate reanalysis data. While the system demonstrated strong predictive performance, its implementation complexity requiring expertise in ERA5 data access and preprocessing, as well as deep learning infrastructure makes it challenging to deploy in resource-limited agricultural environments.

Jhajharia et al. [9] (2025) applied machine learning models, with SVM achieving the best performance, to predict crop yield using satellite and rainfall data. The study demonstrated that publicly available precipitation data can be a strong predictor of crop yield when combined with remote sensing inputs. However, the system output was purely predictive and did not incorporate a decision support layer. The model provided yield estimates but not guidance on how farmers should respond to those estimates.

Islam et al. [10] (2024) developed a climate-aware crop yield prediction system for Saudi Arabia using XGBoost, achieving a remarkably high R^2 value of 0.9745. The study demonstrated the effectiveness of gradient-boosted tree models for regression tasks in agriculture. However, the system was limited to yield prediction and did not provide decision support for irrigation planning or crop management. The geographic focus on Saudi Arabia also limits the generalizability of findings to other climatic regions, particularly the monsoon-dominated agricultural systems of South Asia.

Khaliq et al. [11] (2025) proposed an AI-driven smart agriculture framework that integrates soil analysis, irrigation scheduling, and crop-fertilizer recommendations into a unified decision support platform. This study is perhaps the most closely aligned with the goals of the present project in terms of scope and application. However, the framework relied on direct soil data measurements and did not incorporate multi-temporal climate analysis as a substitute for soil testing. Its dependency on soil analysis infrastructure means it faces the same accessibility challenges as conventional soil testing approaches.

From the above studies, it is observed that most existing systems focus on prediction rather than decision support. Additionally, many approaches rely on expensive infrastructure and do not consider long-term climatic patterns.

Table 2.1 Literature Survey

S.No	Year	Reference	Models & Algorithms	Performance Metrics	Remarks
1	2023	[1]	RF, SVM, Deep Learning	Accuracy ≈ 92.3%	High accuracy but lacks decision support
2	2024	[2]	RF, LSTM	Accuracy ≈ 92%	Limited to short-term forecasting
3	2025	[3]	Regression Trees, Sentinel-2, ERA5	$R^2 = 0.78$	Moderate performance
4	2025	[4]	RF, SVM, KNN	Accuracy improvement ≈ 12%	No long-term analysis
5	2024	[5]	Gradient Boosting, LSTM	Accuracy ≈ 97%	Focused on stress detection
6	2025	[6]	Random Forest, IoT System	Accuracy ≈ 90.1%	High cost due to IoT
7	2024	[7]	Review (AI & Data Analytics)	Conceptual	No specific model; no performance metrics provided
8	2023	[8]	LSTM, XGBoost	Improved performance	Complex implementation
9	2025	[9]	SVM, Satellite Data	Good performance	No decision support
10	2024	[10]	XGBoost	$R^2 = 0.9745$	Prediction-focused
11	2025	[11]	AI Framework	Conceptual improvement	No multi-temporal support

CHAPTER - 3

SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The proposed smart farming system processes multi-temporal climate data and applies machine learning models such as Decision Tree and Gradient Boosting for land suitability classification. Compared to deep learning models, these algorithms require relatively lower computational resources, making the system suitable for deployment in resource-constrained environments.

Efficient hardware is required to handle tasks such as data preprocessing, feature engineering, ACLI computation, model training, and evaluation. A system with a multi-core processor ensures faster data processing and efficient execution of machine learning algorithms.

Since the dataset consists of structured climate data rather than large image or video data, GPU acceleration is not mandatory. However, moderate RAM is required to handle dataset operations and intermediate computations during model training.

Adequate storage is also necessary for storing datasets, trained models, and output results. The overall system is lightweight and can run efficiently on standard computing devices.

3.2 Software Requirements

The software environment is designed to support efficient development, training, and evaluation of machine learning models for agricultural decision support. Python is used as the primary programming language due to its rich ecosystem of data analysis and machine learning libraries. The framework is developed in a reproducible and scalable environment using Anaconda Navigator for package and environment management. Jupyter Notebook provides an interactive platform for data preprocessing, visualization, and model development, while Visual Studio Code (VS Code) can be used for structured coding. Since the system utilizes tree-based machine learning models such as Decision Tree and Gradient Boosting, it does not require GPU-based frameworks and can run efficiently on standard computing systems.

Software Requirements Used in This Project:

- Operating System: Windows / Linux (Ubuntu)
- Development Environment: Anaconda Navigator
- IDE / Notebook: Jupyter Notebook / VS Code

- Programming Language: Python (version 3.8 or above)
- Machine Learning Framework: Scikit-learn

3.2.1 Packages and Libraries

The proposed smart farming system relies on several Python libraries for data preprocessing, machine learning modeling, visualization, and performance evaluation. These libraries enable efficient handling of multi-temporal climate datasets and implementation of classification models.

The major libraries used in this project are listed below:

• NumPy

Used for numerical computations, array manipulation, and efficient handling of climate data.

• Pandas

Used for:

- Data loading
- Preprocessing
- Handling missing values
- Managing structured datasets

• Matplotlib and Seaborn

Used for:

- Data visualization
- Plotting graphs (ACLI trends, predictions)
- Model performance comparison

• Scikit-learn (sklearn)

Used for implementing machine learning models and evaluation.

a) train_test_split

Used for splitting the dataset into training and testing sets without data leakage.

b) MinMaxScaler / StandardScaler

Used for feature scaling and normalization of input data.

c) Machine Learning Models

- Decision Tree Classifier
- Gradient Boosting Classifier

d) Evaluation Metrics

- accuracy_score
- precision_score
- recall_score
- f1_score

• Joblib

Used for saving and loading trained models for future predictions.

• OS and Pathlib

Used for file handling, dataset management, and organizing project workflows.

CHAPTER - 4

DESIGN AND IMPLEMENTATION

4.1 Overview of the Proposed System

This chapter presents a detailed account of the methodology employed in this study for smart farming decision support using multi-temporal climate data. The proposed system is designed as a multi-stage computational pipeline that integrates data preprocessing, feature engineering, agricultural indices computation, and machine learning models. The framework leverages climatic parameters such as rainfall, temperature, soil moisture, and humidity to evaluate agricultural land conditions and provide decision support. The pipeline is structured to enable analysis and comparison of tree-based machine learning models under consistent experimental conditions. Figure 4.1 illustrates the complete system architecture.

The complete system pipeline proceeds as follows. Raw climate data consisting of rainfall, temperature, soil moisture, and humidity values is collected and organized into a structured dataset. The data undergoes preprocessing steps including cleaning, normalization, and handling of missing values. Temporal features such as month and seasonal patterns are incorporated to capture climatic variations. From the processed data, agricultural indices such as Irrigation Index, Crop Index, and Agricultural Condition Level Index (ACLI) are computed to represent environmental conditions.

The dataset is then divided into training and testing subsets. A regression model is used to analyze and predict agricultural indices, capturing continuous relationships between climate variables and land conditions. In parallel, the ACLI values are used to generate categorical labels such as poor, moderate, and good, which are used to train classification models. Machine learning models including Decision Tree and Gradient Boosting are applied for classification, and their performance is evaluated using standard metrics. The best-performing model is selected to generate final predictions and provide recommendations for irrigation planning, crop selection, and seasonal decision-making.

4.2 Dataset Description

4.2.1 Dataset Overview

The dataset used in this study consists of multi-temporal climate data including rainfall, temperature, and soil moisture parameters collected over a period of time. These variables play a significant role in understanding environmental conditions that affect agricultural productivity and land suitability. The dataset is organized in a structured format, where each record corresponds to a specific time period (monthly basis), enabling the analysis of temporal variations in climatic conditions.

The input features include Root Zone Soil Wetness (RZSW), Surface Soil Wetness (SSW), precipitation (rainfall), and temperature, along with temporal attributes such as month and season. Instead of a direct target variable, the Agricultural Condition Level Index (ACLI) is computed using these features to represent the overall condition of agricultural land.
Based on ACLI values, the land is categorized into three classes: poor, moderate, and good. This dataset structure allows the system to capture both short-term variations and long-term seasonal trends, which are essential for effective agricultural decision support.

4.2.2 Data Preprocessing

Before model training, the dataset undergoes a comprehensive preprocessing pipeline to ensure data quality and consistency. Data integration is performed to combine different climate variables into a unified dataset. Missing values and inconsistencies are handled using statistical methods such as mean or median imputation to maintain data reliability. Data cleaning is applied to remove noise, inconsistencies, and outliers that may affect model performance. Temporal alignment is ensured so that all climate variables correspond to the same time index.

To ensure uniform scaling of features, Min-Max normalization is applied to all numerical attributes, defined as:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This scaling ensures that all features lie within a uniform range, improving model performance, stability, and convergence during training. The preprocessed dataset is then used for feature engineering, agricultural indices computation, and subsequent model development.

4.2.3 Feature Engineering

Feature engineering is performed to extract meaningful information from raw climate data and improve model performance. Temporal features such as month and seasonal indicators are derived to capture climatic variations over time. Key input features include rainfall, temperature, relative humidity, and root zone soil moisture, which represent environmental and soil conditions. Based on these features, agricultural indices such as Irrigation Index, Crop Index, and Agricultural Condition Level Index (ACLI) are computed to summarize overall land suitability. These features and derived indices help the model capture seasonal patterns and improve classification accuracy.

4.2.4 Train-Test Data Splitting

The dataset is divided into training and testing subsets using an 80–20 split strategy. The data is split without shuffling to preserve the temporal order of climate observations, ensuring that past data is used for training and future data is used for testing. This approach prevents data leakage and provides a realistic evaluation of model performance.

4.3 Overall System Architecture

Figure 4.1 presents the complete multi-stage pipeline of the proposed smart farming decision support system. The architecture begins with climate data input, followed by preprocessing and feature engineering. The processed data is used to compute agricultural indices, including the Agricultural Condition Level Index (ACLI), and is then divided into training and testing datasets. Machine learning models such as Decision Tree and Gradient Boosting are applied for land suitability classification. Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. Finally, the system provides recommendations for irrigation planning, crop selection, and seasonal decision-making.

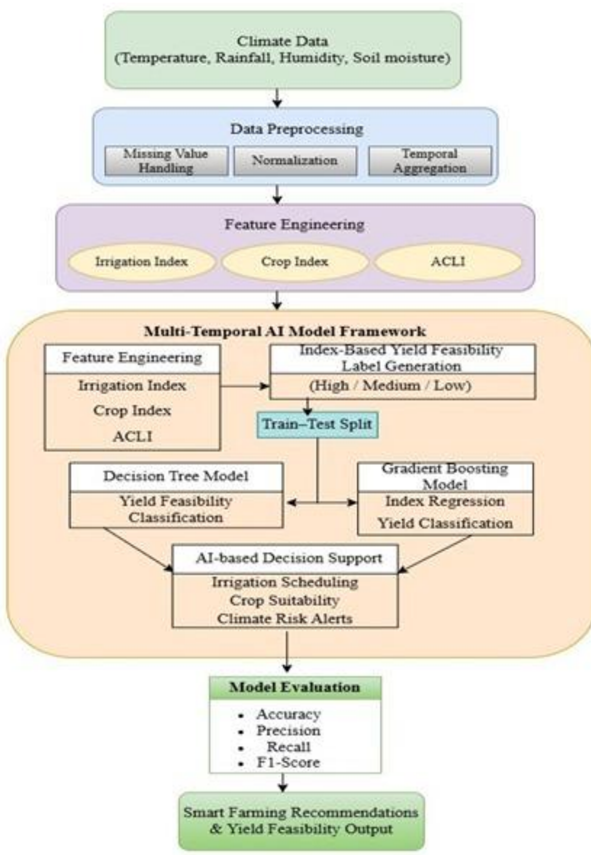

Figure 4.1: Proposed Model for Smart Farming Decision Support System using Multi-Temporal Climate Data

4.4 Model Architectures

This section presents the machine learning models employed for agricultural land suitability classification using multi-temporal climate data. The models are designed to analyze relationships between climatic variables such as rainfall, temperature, soil moisture, and humidity, and their impact on agricultural conditions. Initially, agricultural indices such as Irrigation Index, Crop Index, and Agricultural Condition Level Index (ACLI) are generated from the input features. These indices are further used for analysis and classification.

Two tree-based machine learning models are implemented, namely Decision Tree and Gradient Boosting. These models are selected due to their ability to handle structured data, provide interpretability, and capture non-linear relationships effectively. A regression model is used to analyze and predict agricultural indices, while classification models are used to categorize land suitability based on ACLI values.

Each model follows a structured pipeline consisting of data input, feature processing, model training, and prediction. The Decision Tree model provides interpretable rule-based classification, while the Gradient Boosting model improves accuracy by combining multiple trees through ensemble learning. The following subsections provide a detailed description of each model and their working principles.

4.4.1 Model 1: Decision Tree

The Decision Tree is a supervised machine learning algorithm used for classification tasks. It is selected as a baseline model due to its simplicity, interpretability, and ability to handle structured data effectively. The model works by splitting the dataset into subsets based on feature values, forming a tree-like structure of decisions.

The workflow of the Decision Tree model begins with input features such as rainfall, temperature, soil moisture, and humidity, along with the computed agricultural indices, particularly the Agricultural Condition Level Index (ACLI). The model recursively splits the data based on the most informative feature at each node using criteria such as entropy and information gain. This process continues until the data is classified into categories such as poor, moderate, and good land conditions.
The mathematical representation is based on entropy and information gain:
Entropy:

$$H(S) = -\sum p_i \log_2(p_i)$$

Information Gain:

$$IG(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

The Decision Tree model generates rule-based decisions that are easy to interpret. However, it may not effectively capture complex relationships between variables, which motivates the use of more advanced ensemble models such as Gradient Boosting. The workflow of this model is illustrated in Figure 4.2.

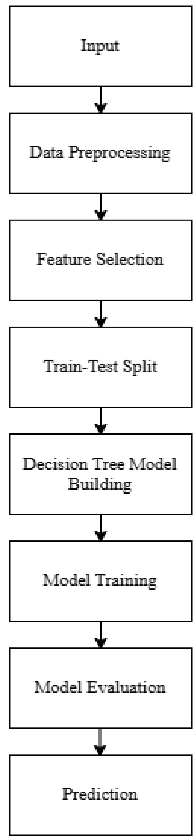

Figure 4.2: Decision Tree Model Workflow

4.4.2 Model 2: Gradient Boosting

Gradient Boosting is an ensemble machine learning technique that improves prediction accuracy by combining multiple weak learners, typically decision trees. Unlike a single Decision Tree, Gradient Boosting builds models sequentially, where each new tree learns from the errors of the previous ones. This enables the model to capture complex non-linear relationships between climate variables and agricultural conditions.

In this study, the model utilizes derived agricultural indices such as the Agricultural Condition Level Index (ACLI), along with climate features including rainfall, temperature, soil moisture, and humidity. The model iteratively updates predictions by minimizing a loss function, resulting in improved classification performance.
The mathematical representation of Gradient Boosting is given by:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

Where:

- $F_m(x)$→ final model at stage m
- $h_m(x)$→ weak learner (decision tree)
- η→ learning rate

The model is trained by sequentially adding trees and optimizing parameters such as learning rate and the number of estimators. Compared to the Decision Tree model, Gradient Boosting provides higher accuracy and better generalization.

Due to its ability to handle complex patterns and reduce prediction errors, Gradient Boosting is selected as the final model for land suitability classification. The workflow of this model is illustrated in Figure 4.3.

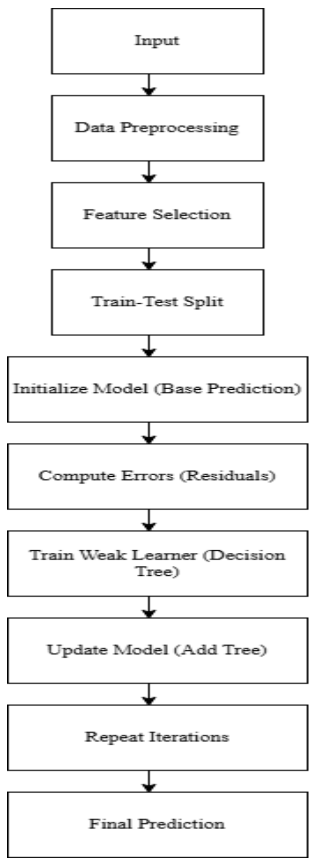

Figure 4.3: Gradient Boosting Model Workflow

4.5 Training Strategy

A consistent and well-defined training strategy is applied to ensure reliable model performance and reproducibility. The dataset is divided into training and testing subsets using an 80–20 split without shuffling to preserve the temporal order of climate data. All input features are normalized using Min-Max scaling to ensure uniform feature ranges and improve model performance.

The training process involves both regression and classification models. A Decision Tree Regressor is used to analyze and predict agricultural indices such as Irrigation Index, Crop Index, and ACLI. For classification, Decision Tree and Gradient Boosting models are trained to categorize land suitability into poor, moderate, and good classes based on ACLI values. Hyperparameters such as maximum depth, minimum samples split, number of estimators, and learning rate are selected based on empirical tuning to achieve optimal performance. The models are trained using the training dataset and evaluated on the testing dataset to ensure generalization.

To ensure reproducibility, random states are fixed during model initialization. Overfitting is controlled by limiting tree depth and tuning model parameters appropriately. The overall training strategy ensures efficient learning, stable performance, and accurate classification of agricultural land suitability.

4.5 Experimental Workflow Summary

The complete experimental pipeline integrates all components described in the preceding sections into a structured and reproducible workflow for smart farming decision support. The proposed framework systematically processes multi-temporal climate data through preprocessing, feature engineering, agricultural indices computation, and machine learning-based analysis. The sequential steps of the experimental procedure are as follows.

Step 1 - Dataset Acquisition:
The climate dataset comprising rainfall, temperature, relative humidity, and soil moisture variables is collected and organized into a structured format. Each record is indexed based on time, and relevant features are extracted to form a unified dataset. The dataset is verified for consistency before preprocessing.

Step 2 -Data Preprocessing:
The dataset undergoes preprocessing steps including data cleaning, missing value handling, and normalization. Temporal alignment is ensured across all variables. Feature scaling is performed using Min-Max normalization, and temporal attributes such as month and seasonal indicators are incorporated. The dataset is then split into training and testing subsets using an 80–20 ratio without shuffling to preserve temporal order.

Step 3 -Feature Engineering:

Relevant features such as rainfall, temperature, humidity, and soil moisture are selected. Monthly aggregated features are generated using rolling window techniques to capture temporal trends in climate data.

Step 4 - Agricultural Indices Computation:
Agricultural indices including Irrigation Index, Crop Index, and Agricultural Condition Level Index (ACLI) are computed using normalized climate features. These indices summarize environmental conditions and serve as key inputs for model development.

Step 5 - Label Generation:
The ACLI values are categorized into three classes—poor, moderate, and good—based on predefined thresholds. These labels represent land suitability and are used for classification.

Step 6 - Regression Model Training:
A Decision Tree Regressor is trained to analyze and predict agricultural indices, capturing continuous relationships between climate variables and land conditions.

Step 7 - Classification Model Training:
Decision Tree and Gradient Boosting classifiers are trained using the labeled dataset to classify land suitability. These models learn patterns from the input features and indices to generate accurate predictions.

Step 8 - Model Evaluation:
The trained models are evaluated on the testing dataset using performance metrics such as accuracy, precision, recall, and F1-score. A comparative analysis is performed to identify the best-performing model.

Step 9 - Decision Support Output:
The final model is used to generate predictions and provide recommendations for irrigation planning, crop selection, and seasonal decision-making.

This structured experimental workflow ensures consistent data processing, model training, and evaluation, enabling accurate and reliable smart farming decision support.

CHAPTER - 5

SOURCE CODE

5.1 Source Code For Decision Tree Workflow

```
from google.colab import drive
drive.mount('/content/drive/')
import pandas as pd
import numpy as np
df = pd.read_csv('/content/drive/MyDrive/Daily.csv')
df['Date'] = pd.to_datetime(df['YEAR'].astype(str) + ' ' + df['DOY'].astype(str),
format='%Y %j')
df = df.set_index('Date')
df = df.sort_index()
df.replace(-999, np.nan, inplace=True)
print(df.head())
for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        if df[column].isnull().any():
            df[column].fillna(df[column].mean(), inplace=True)
print("Missing values after imputation:")
print(df.isnull().sum())
for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        if df[column].isnull().any():
            df[column] = df[column].fillna(df[column].mean())
print("Missing values after imputation:")
print(df.isnull().sum())
df['monthly_cumulative_rainfall'] = df['PRECTOTCORR'].rolling(window=30).sum()
df['monthly_average_temperature'] = df['T2M'].rolling(window=30).mean()
```

```
df['monthly_average_relative_humidity'] = df['RH2M'].rolling(window=30).mean()
df['monthly_average_root_zone_soil_moisture'] =
df['GWETROOT'].rolling(window=30).mean()
df.dropna(inplace=True)
print("DataFrame with new monthly features and NaNs dropped:")
print(df.head())
from sklearn.preprocessing import MinMaxScaler
monthly_features = [
    'monthly_cumulative_rainfall',
    'monthly_average_temperature',
    'monthly_average_relative_humidity',
    'monthly_average_root_zone_soil_moisture'
]
scaler = MinMaxScaler()
df[monthly_features] = scaler.fit_transform(df[monthly_features])
print("DataFrame with normalized monthly features:")
print(df[monthly_features].head())
df['Irrigation Index'] = (df['monthly_average_root_zone_soil_moisture'] +
df['monthly_cumulative_rainfall']) / 2
df['Crop Index'] = (df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 3
df['ACLI'] = (df['monthly_cumulative_rainfall'] + df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 4
print("DataFrame with new agricultural indices:")
print(df[['Irrigation Index', 'Crop Index', 'ACLI']].head())
lower_bound = df['ACLI'].quantile(0.33)
upper_bound = df['ACLI'].quantile(0.66)

def assign_yield_feasibility(acli):
```

```
    if acli <= lower_bound:
        return 'Low'
    elif acli >= upper_bound:
        return 'High'
    else:
        return 'Medium'
df['Yield Feasibility'] = df['ACLI'].apply(assign_yield_feasibility)
print("Yield Feasibility categories and their counts:")
print(df['Yield Feasibility'].value_counts())
monthly_features = [
    'monthly_cumulative_rainfall',
    'monthly_average_temperature',
    'monthly_average_relative_humidity',
    'monthly_average_root_zone_soil_moisture'
]
X = df[monthly_features]
y_regression = df[['Irrigation Index', 'Crop Index', 'ACLI']]
y_classification = df['Yield Feasibility']
split_point = int(len(df) * 0.8)
X_train = X.iloc[:split_point]
X_test = X.iloc[split_point:]
y_regression_train = y_regression.iloc[:split_point]
y_regression_test = y_regression.iloc[split_point:]
y_classification_train = y_classification.iloc[:split_point]
y_classification_test = y_classification.iloc[split_point:]
print("Shapes of the datasets:")
print(f"X_train: {X_train.shape}")
print(f"X_test: {X_test.shape}")
print(f"y_regression_train: {y_regression_train.shape}")
print(f"y_regression_test: {y_regression_test.shape}")
print(f"y_classification_train: {y_classification_train.shape}")
```

```
print(f"y_classification_test: {y_classification_test.shape}")
from sklearn.tree import DecisionTreeRegressor
import joblib
dt_model = DecisionTreeRegressor(
    random_state=42,
    max_depth=10
)
dt_model.fit(X_train, y_regression_train)
joblib.dump(dt_model, "regression_model.pkl")
print("Decision Tree Regressor trained and saved.")
from sklearn.tree import DecisionTreeClassifier
import joblib
classification_model = DecisionTreeClassifier(
    random_state=42,
    max_depth=10,
    min_samples_split=5
)
classification_model.fit(X_train, y_classification_train)
joblib.dump(classification_model, "classification_model.pkl")
print("Decision Tree Classifier trained and saved successfully.")
import joblib
regression_model = joblib.load('regression_model.pkl')
print("Regression model loaded successfully.")
classification_model = joblib.load('classification_model.pkl')
print("Classification model loaded successfully.")
print("\nShapes of test datasets for evaluation:")
print(f"X_test: {X_test.shape}")
print(f"y_regression_test: {y_regression_test.shape}")
print(f"y_classification_test: {y_classification_test.shape}")
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np
```

```
y_regression_pred = regression_model.predict(X_test)
y_regression_pred_df = pd.DataFrame(y_regression_pred,
columns=y_regression_test.columns, index=y_regression_test.index)
print("Regression Model Evaluation:")
for col in y_regression_test.columns:
    mae = mean_absolute_error(y_regression_test[col], y_regression_pred_df[col])
    rmse = np.sqrt(mean_squared_error(y_regression_test[col], y_regression_pred_df[col]))
    r2 = r2_score(y_regression_test[col], y_regression_pred_df[col])

    print(f"\n{col}:")
    print(f"  MAE: {mae:.4f}")
    print(f"  RMSE: {rmse:.4f}")
    print(f"  R^2: {r2:.4f}")
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_regression_test['ACLI'], y=y_regression_pred_df['ACLI'])
plt.xlabel('Actual ACLI')
plt.ylabel('Predicted ACLI')
plt.title('Actual vs. Predicted ACLI Values')
plt.plot([y_regression_test['ACLI'].min(), y_regression_test['ACLI'].max()],
         [y_regression_test['ACLI'].min(), y_regression_test['ACLI'].max()],
         'r--')
plt.grid(True)
plt.show()
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
classification_report
y_classification_pred = classification_model.predict(X_test)
print("Classification Model Evaluation:")
accuracy = accuracy_score(y_classification_test, y_classification_pred)
print(f"Accuracy: {accuracy:.4f}")
```

```
print("\nClassification Report:")
print(classification_report(y_classification_test, y_classification_pred))
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix
classes = sorted(y_classification_test.unique())
cm = confusion_matrix(y_classification_test, y_classification_pred, labels=classes)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=classes,
yticklabels=classes)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix for Yield Feasibility Classification')
plt.show()
```

5.2 Source Code For Gradient Boosting Workflow

```
from google.colab import drive
drive.mount('/content/drive/')
import pandas as pd
import numpy as np
df = pd.read_csv('/content/drive/MyDrive/Daily.csv')
df['Date'] = pd.to_datetime(df['YEAR'].astype(str) + ' ' + df['DOY'].astype(str),
format='%Y %j')
df = df.set_index('Date')
df = df.sort_index()
df.replace(-999, np.nan, inplace=True)
print("DataFrame reloaded, 'Date' column created and set as index, sorted, and -999
replaced with NaN.")
print(df.head())
for column in df.columns:
   if df[column].dtype in ['int64', 'float64']:
```

```
        if df[column].isnull().any():
            df[column].fillna(df[column].mean(), inplace=True)
print("Missing values after imputation:")
print(df.isnull().sum())

for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        if df[column].isnull().any():
            df[column] = df[column].fillna(df[column].mean())
print("Missing values after imputation:")
print(df.isnull().sum())

df['monthly_cumulative_rainfall'] = df['PRECTOTCORR'].rolling(window=30).sum()
df['monthly_average_temperature'] = df['T2M'].rolling(window=30).mean()
df['monthly_average_relative_humidity'] = df['RH2M'].rolling(window=30).mean()
df['monthly_average_root_zone_soil_moisture'] =
df['GWETROOT'].rolling(window=30).mean()
df.dropna(inplace=True)
print("DataFrame with new monthly features and NaNs dropped:")
print(df.head())

from sklearn.preprocessing import MinMaxScaler
monthly_features = [
    'monthly_cumulative_rainfall',
    'monthly_average_temperature',
    'monthly_average_relative_humidity',
    'monthly_average_root_zone_soil_moisture'
]
scaler = MinMaxScaler()
df[monthly_features] = scaler.fit_transform(df[monthly_features])
print("DataFrame with normalized monthly features:")
```

```
print(df[monthly_features].head())

df['Irrigation Index'] = (df['monthly_average_root_zone_soil_moisture'] +
df['monthly_cumulative_rainfall']) / 2
df['Crop Index'] = (df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 3
df['ACLI'] = (df['monthly_cumulative_rainfall'] + df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 4
print("DataFrame with new agricultural indices:")
print(df[['Irrigation Index', 'Crop Index', 'ACLI']].head())
lower_bound = df['ACLI'].quantile(0.33)
upper_bound = df['ACLI'].quantile(0.66)
def assign_yield_feasibility(acli):
    if acli <= lower_bound:
        return 'Low'
    elif acli >= upper_bound:
        return 'High'
    else:
        return 'Medium'
df['Yield Feasibility'] = df['ACLI'].apply(assign_yield_feasibility)
print("Yield Feasibility categories and their counts:")
print(df['Yield Feasibility'].value_counts())
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.ensemble import GradientBoostingRegressor
import joblib
from google.colab import drive
drive.mount('/content/drive/')
```

```
df = pd.read_csv('/content/drive/MyDrive/Daily.csv')
df['Date'] = pd.to_datetime(df['YEAR'].astype(str) + ' ' + df['DOY'].astype(str),
format='%Y %j')
df = df.set_index('Date')
df = df.sort_index()
df.replace(-999, np.nan, inplace=True)
for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        if df[column].isnull().any():
            df[column] = df[column].fillna(df[column].mean())
df['monthly_cumulative_rainfall'] = df['PRECTOTCORR'].rolling(window=30).sum()
df['monthly_average_temperature'] = df['T2M'].rolling(window=30).mean()
df['monthly_average_relative_humidity'] = df['RH2M'].rolling(window=30).mean()
df['monthly_average_root_zone_soil_moisture'] =
df['GWETROOT'].rolling(window=30).mean()
df.dropna(inplace=True)
monthly_features = [
    'monthly_cumulative_rainfall',
    'monthly_average_temperature',
    'monthly_average_relative_humidity',
    'monthly_average_root_zone_soil_moisture'
]
scaler = MinMaxScaler()
df[monthly_features] = scaler.fit_transform(df[monthly_features])
df['Irrigation Index'] = (df['monthly_average_root_zone_soil_moisture'] +
df['monthly_cumulative_rainfall']) / 2
df['Crop Index'] = (df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 3
```

```
df['ACLI'] = (df['monthly_cumulative_rainfall'] + df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 4
X = df[monthly_features]
y_regression = df[['Irrigation Index', 'Crop Index', 'ACLI']]
split_point = int(len(df) * 0.8)
X_train = X.iloc[:split_point]
y_regression_train = y_regression.iloc[:split_point]
target_columns = ['Irrigation Index', 'Crop Index', 'ACLI']
for col in target_columns:
    print(f"Training model for {col}...")
    regression_model = GradientBoostingRegressor(
        n_estimators=200,
        learning_rate=0.05,
        random_state=42
    )
    regression_model.fit(X_train, y_regression_train[col])
    filename = f"/content/drive/MyDrive/{col.lower().replace(' ', '_')}_regressor.pkl"
    joblib.dump(regression_model, filename)
    print(f"Gradient Boosting Regressor for {col} trained and saved as {filename}.")
from sklearn.ensemble import GradientBoostingClassifier
import joblib
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from google.colab import drive
drive.mount('/content/drive/')
df = pd.read_csv('/content/drive/MyDrive/Daily.csv')
df['Date'] = pd.to_datetime(df['YEAR'].astype(str) + ' ' + df['DOY'].astype(str),
format='%Y %j')
df = df.set_index('Date')
```

```
df = df.sort_index()
df.replace(-999, np.nan, inplace=True)
for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        if df[column].isnull().any():
            df[column] = df[column].fillna(df[column].mean())
df['monthly_cumulative_rainfall'] = df['PRECTOTCORR'].rolling(window=30).sum()
df['monthly_average_temperature'] = df['T2M'].rolling(window=30).mean()
df['monthly_average_relative_humidity'] = df['RH2M'].rolling(window=30).mean()
df['monthly_average_root_zone_soil_moisture'] =
df['GWETROOT'].rolling(window=30).mean()
df.dropna(inplace=True)
monthly_features = [
    'monthly_cumulative_rainfall',
    'monthly_average_temperature',
    'monthly_average_relative_humidity',
    'monthly_average_root_zone_soil_moisture'
]
scaler = MinMaxScaler()
df[monthly_features] = scaler.fit_transform(df[monthly_features])
df['Irrigation Index'] = (df['monthly_average_root_zone_soil_moisture'] +
df['monthly_cumulative_rainfall']) / 2
df['Crop Index'] = (df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 3
df['ACLI'] = (df['monthly_cumulative_rainfall'] + df['monthly_average_temperature'] +
df['monthly_average_relative_humidity'] +
df['monthly_average_root_zone_soil_moisture']) / 4
lower_bound = df['ACLI'].quantile(0.33)
upper_bound = df['ACLI'].quantile(0.66)
def assign_yield_feasibility(acli):
```

```
    if acli <= lower_bound:
        return 'Low'
    elif acli >= upper_bound:
        return 'High'
    else:
        return 'Medium'
df['Yield Feasibility'] = df['ACLI'].apply(assign_yield_feasibility)
X = df[monthly_features]
y_classification = df['Yield Feasibility']
split_point = int(len(df) * 0.8)
X_train = X.iloc[:split_point]
X_test = X.iloc[split_point:]
y_classification_train = y_classification.iloc[:split_point]
y_classification_test = y_classification.iloc[split_point:]
classification_model = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
    random_state=42
)
classification_model.fit(X_train, y_classification_train)
joblib.dump(classification_model,
"/content/drive/MyDrive/gradient_boosting_classifier.pkl")
print("Gradient Boosting Classifier trained and saved successfully.")
print(f"X_test shape: {X_test.shape}")
print(f"y_classification_test shape: {y_classification_test.shape}")
import joblib
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
y_regression = df[['Irrigation Index', 'Crop Index', 'ACLI']]
```

```
split_point = int(len(df) * 0.8)
y_regression_test = y_regression.iloc[split_point:]
irrigation_model = joblib.load('/content/drive/MyDrive/irrigation_index_regressor.pkl')
crop_model = joblib.load('/content/drive/MyDrive/crop_index_regressor.pkl')
acli_model = joblib.load('/content/drive/MyDrive/acli_regressor.pkl')
print("Regression models loaded successfully.")
y_pred_irrigation = irrigation_model.predict(X_test)
y_pred_crop = crop_model.predict(X_test)
y_pred_acli = acli_model.predict(X_test)
y_regression_pred_df = pd.DataFrame({
    'Irrigation Index': y_pred_irrigation,
    'Crop Index': y_pred_crop,
    'ACLI': y_pred_acli
}, index=X_test.index)
print("\nRegression Model Evaluation:")
for col in y_regression_test.columns:
    mae = mean_absolute_error(y_regression_test[col], y_regression_pred_df[col])
    rmse = np.sqrt(mean_squared_error(y_regression_test[col], y_regression_pred_df[col]))
    r2 = r2_score(y_regression_test[col], y_regression_pred_df[col])
    print(f"\n{col}:")
    print(f"  MAE: {mae:.4f}")
    print(f"  RMSE: {rmse:.4f}")
    print(f"  R^2: {r2:.4f}")
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_regression_test['ACLI'], y=y_regression_pred_df['ACLI'])
plt.xlabel('Actual ACLI')
plt.ylabel('Predicted ACLI')
plt.title('Actual vs. Predicted ACLI Values')
plt.plot([y_regression_test['ACLI'].min(), y_regression_test['ACLI'].max()],
         [y_regression_test['ACLI'].min(), y_regression_test['ACLI'].max()],
         'r--') # Add a diagonal line for perfect prediction
```

```
plt.grid(True)
plt.show()
import joblib
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
classification_model =
joblib.load('/content/drive/MyDrive/gradient_boosting_classifier.pkl')
print("Gradient Boosting Classifier loaded successfully.")
y_classification_pred = classification_model.predict(X_test)
print("\nClassification Model Evaluation:")
accuracy = accuracy_score(y_classification_test, y_classification_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_classification_test, y_classification_pred))
classes = sorted(y_classification_test.unique())
cm = confusion_matrix(y_classification_test, y_classification_pred, labels=classes)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=classes,
yticklabels=classes)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix for Yield Feasibility Classification')
plt.show()
```

CHAPTER – 6

RESULT ANALYSIS AND DISCUSSION

6.1 Performance Metrics: Theory and Formulae

The evaluation of machine learning models in this study requires a combination of regression and classification performance metrics to assess both prediction accuracy and decision-making capability. Regression metrics are used to evaluate the prediction of agricultural indices, while classification metrics are used to assess land suitability categorization. Therefore, multiple complementary metrics are employed to provide a comprehensive evaluation of model performance.

6.1.1 Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is used to evaluate regression performance by measuring the square root of the average squared differences between predicted and actual values. It penalizes larger errors more heavily and is useful for assessing the accuracy of predicted agricultural indices such as ACLI.

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

where $y_i$represents the actual value, $\hat{y}_i$represents the predicted value, and $n$is the number of observations. A lower RMSE indicates better model performance.

6.1.2 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values. It treats all errors equally and provides a clear interpretation of prediction accuracy.

$$MAE = \frac{1}{n}\sum \mid y_i - \hat{y}_i \mid$$

MAE is useful for understanding the average deviation in predicted agricultural indices.

6.1.3 Coefficient of Determination (R^2 Score)

The R^2 Score measures how well the regression model explains the variance in the target variable.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

A value closer to 1 indicates better model performance and stronger predictive capability.

6.1.4 Classification Metrics

In addition to regression evaluation, classification metrics are used to assess the performance of land suitability classification.

1. **Accuracy**
   Accuracy measures the proportion of correctly classified instances:
   $$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2. **Precision**
   Precision measures the proportion of correctly predicted positive instances:
   $$Precision = \frac{TP}{TP + FP}$$

3. **Recall**
   Recall measures the ability of the model to identify all relevant instances:
   $$Recall = \frac{TP}{TP + FN}$$

4. **F1-Score**
   F1-Score is the harmonic mean of precision and recall:
   $$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

6.1.5 Overall Evaluation

The combination of regression and classification metrics provides a comprehensive evaluation of the proposed system. Regression metrics assess the accuracy of predicted agricultural indices, while classification metrics evaluate the effectiveness of land suitability prediction. Together, these metrics ensure that the model performs well both in continuous prediction and practical decision-making.

6.2 Comprehensive Performance Metrics

Table 6.1 and Table 6.2 present the complete set of performance metrics for all models evaluated in this study, including both regression and classification tasks. The models are tested on a held-out 20% dataset to ensure unbiased evaluation.

The regression models (Decision Tree Regressor and Gradient Boosting Regressor) are evaluated using RMSE, MAE, and R^2 Score, while the classification models (Decision Tree Classifier and Gradient Boosting Classifier) are evaluated using Accuracy, Precision, Recall, and F1-score.

All results are obtained under identical preprocessing, feature engineering, and experimental conditions to ensure a fair comparison.

Model	**Target Variable**	**MAE**	**RMSE**	**R^2 Score**
Decision Tree	Irrigation Index	0.0244	0.0350	0.9824
Decision Tree	Crop Index	0.0290	0.0483	0.9435
Decision Tree	Agricultural Condition Level Index	0.0250	0.0414	0.9622
Gradient Boosting	Irrigation Index	0.0107	0.0152	0.9967
Gradient Boosting	Crop Index	0.0166	0.0238	0.9863
Gradient Boosting	Agricultural Condition Level Index	0.0165	0.0238	0.9875

Table 6.1 – Regression Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Decision Tree Classifier	0.9048	0.91	0.90	0.91
Gradient Boosting Classifier	0.9261	0.93	0.93	0.93

Table 6.2 – Classification Performance Metrics

The regression results indicate that the Gradient Boosting model significantly outperforms the Decision Tree model across all target variables. It achieves lower MAE and RMSE values along with higher R^2 scores, demonstrating superior prediction accuracy and reliability.

In terms of classification, the Gradient Boosting classifier achieves higher accuracy (92.61%) compared to the Decision Tree classifier (90.48%). Additionally, improvements in precision, recall, and F1-score indicate better class-wise balance and generalization capability.
Overall, the results confirm that Gradient Boosting is more effective in capturing complex nonlinear relationships in climate-based agricultural data.

6.2 Training and Evaluation Graphs

6.2.1 Regression Evaluation — Actual vs Predicted ACLI

The performance of the regression models is evaluated using Actual vs Predicted ACLI plots for both the Decision Tree and Gradient Boosting models. These plots provide a visual representation of how closely the predicted values match the actual values.
In these plots, the x-axis represents the actual ACLI values, while the y-axis represents the predicted ACLI values. A diagonal reference line is included to indicate perfect prediction, where predicted values exactly match the actual values.

Figure 6.1 illustrates the performance of the Decision Tree Regressor. It can be observed that the predicted values generally follow the diagonal reference line, indicating that the model captures the relationship between input features and ACLI reasonably well. However, some deviations are visible, particularly in the mid-range values, which may be due to overfitting and limited generalization capability of the model.

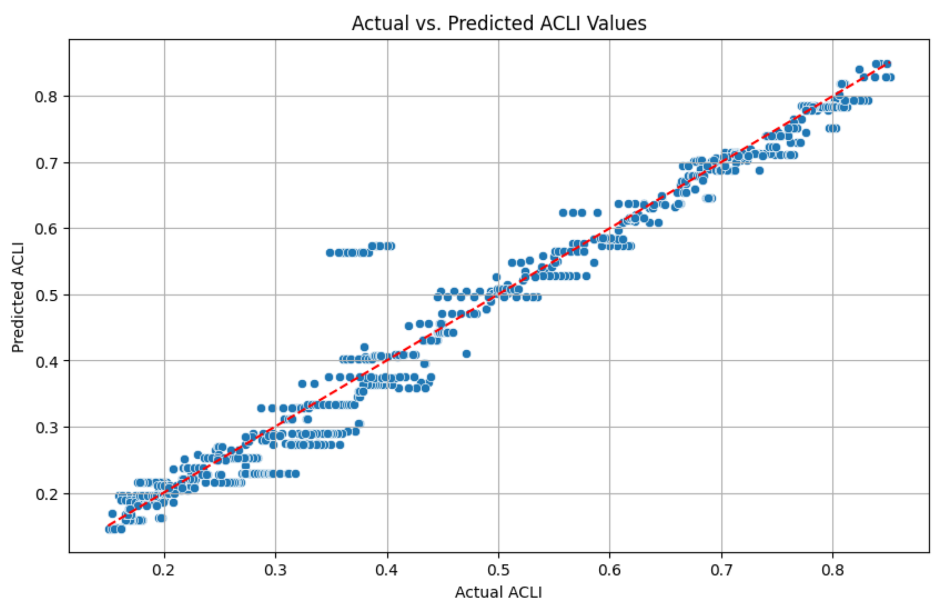

Fig- 6.1: Actual vs Predicted ACLI Values – Decision Tree Regressor

Figure 6.2 illustrates the performance of the Gradient Boosting Regressor. In comparison, the data points are more tightly clustered around the reference line, indicating improved prediction accuracy. The model demonstrates better generalization and reduced error due to its ensemble learning approach, where successive models correct the errors of previous ones.

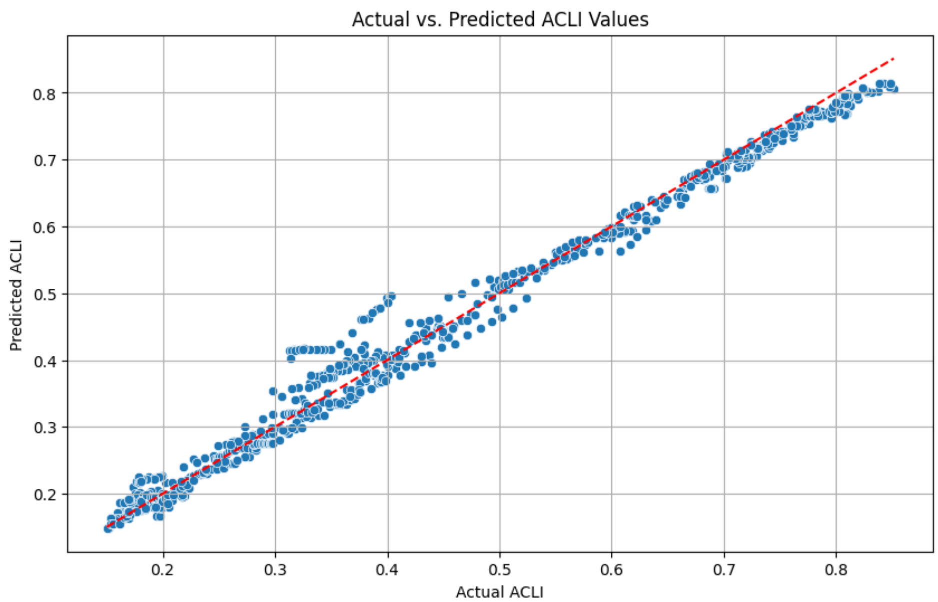

Fig- 6.2: Actual vs Predicted ACLI Values – Gradient Boosting Regressor

The comparative analysis of both figures clearly shows that the Gradient Boosting model provides more accurate and consistent predictions than the Decision Tree model. The reduced spread of data points in the Gradient Boosting plot confirms lower prediction error, which is consistent with the lower MAE and RMSE values and higher R^2 scores observed in Table 6.1.

Overall, both models perform effectively in predicting ACLI values; however, the Gradient Boosting model outperforms the Decision Tree model in terms of accuracy, reliability, and generalization capability.

6.2.2 Classification Evaluation – Confusion Matrix

The performance of the classification models is evaluated using confusion matrices for both the Decision Tree Classifier and the Gradient Boosting Classifier. The confusion matrix provides a detailed comparison between actual and predicted class labels, enabling a deeper understanding of model performance across different classes.

In the confusion matrix, the rows represent the actual classes, while the columns represent the predicted classes. The diagonal elements indicate correctly classified instances, whereas the off-diagonal elements represent misclassifications.

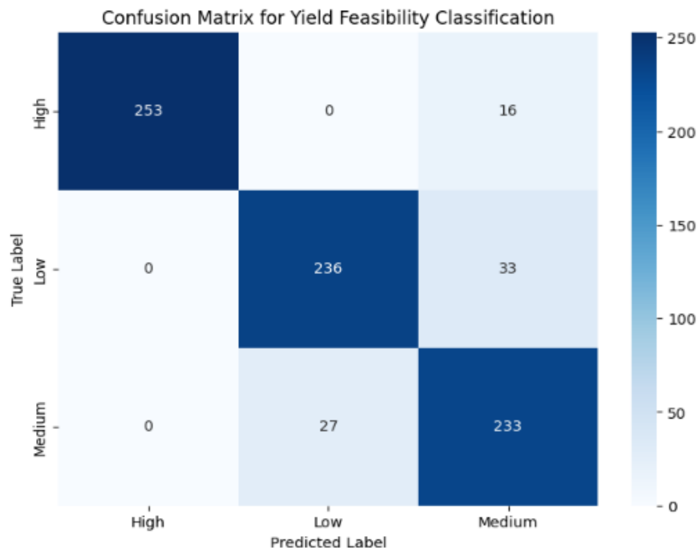

Fig- 6.3: Confusion Matrix of Decision Tree Classifier on ACLI

Figure 6.3 illustrates the confusion matrix of the Decision Tree Classifier. It can be observed that the majority of predictions lie along the diagonal, indicating correct classification. The model performs particularly well in predicting the High class. However, some misclassification is observed between the Medium and Low classes, which may be due to overlapping climatic conditions affecting agricultural suitability.

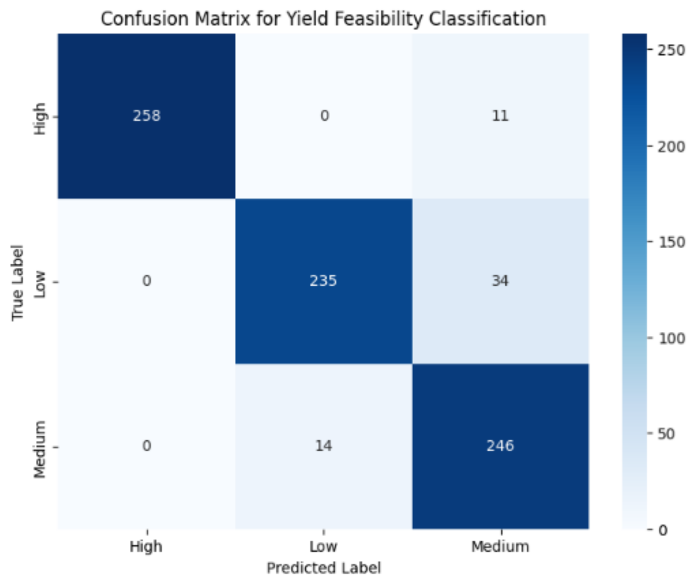

Fig- 6.4: Confusion Matrix of Gradient Boosting Classifier on ACLI

Figure 6.4 illustrates the confusion matrix of the Gradient Boosting Classifier. Compared to the Decision Tree model, the Gradient Boosting model shows a higher concentration of values along the diagonal, indicating improved classification accuracy. The number of misclassifications is reduced, especially between the Medium and Low classes, demonstrating better class separation and generalization.

The comparative analysis of both confusion matrices shows that the Gradient Boosting Classifier provides more accurate and balanced predictions across all classes. This improvement is consistent with the higher accuracy, precision, recall, and F1-score values reported in Table 6.2.

CHAPTER – 7

CONCLUSION

This project has presented a comprehensive, data-driven Smart Farming Decision Support System based on Multi-Temporal Climate Analysis. The system was designed with a clear and consistent objective: to provide practical, accurate, and accessible agricultural decision support to resource-constrained farmers using freely available climate data, without reliance on expensive soil testing laboratories, IoT sensor networks, or satellite imagery processing infrastructure. The experimental results demonstrate that this objective has been successfully achieved.

The core methodological contribution of this work is the introduction of the Agricultural Condition Level Index (ACLI) as a composite climate-based measure of agricultural land suitability. By aggregating four normalized climate parameters monthly cumulative rainfall, average temperature, average relative humidity, and average root zone soil moisture into a single score, the ACLI provides a meaningful and interpretable summary of the environmental conditions affecting crop growth. The ACLI-based threshold mapping scheme generates balanced three-class labels (Low, Medium, High) that directly correspond to Poor, Moderate, and Good land conditions, enabling supervised machine learning-based classification.

The comparative evaluation of two tree-based machine learning models Decision Tree and Gradient Boosting provides strong empirical evidence that Gradient Boosting is the superior choice for this application. The Gradient Boosting Classifier achieved an overall accuracy of 92.61% compared to 90.48% for the Decision Tree. The performance advantage is even more pronounced in the regression task: Gradient Boosting achieves an ACLI R^2 of 0.9875 with RMSE of 0.0238, compared to the Decision Tree's R^2 of 0.9622 and RMSE of 0.0414. These results demonstrate that the ensemble learning approach of Gradient Boosting, which iteratively refines predictions by targeting the residuals of the current ensemble, is well-suited to capturing the nonlinear interactions between climate variables and agricultural condition indices.

Seasonal analysis of the classification results confirms the expected agricultural dynamics of the study region. Monsoon season records are predominantly classified as High (Good), summer records as Low (Poor), and transitional seasons as Medium (Moderate). This seasonal consistency validates the ecological relevance of the ACLI framework and demonstrates that the system is capturing genuine climatic patterns rather than statistical artefacts. The ability to anticipate these seasonal shifts enables farmers to plan irrigation schedules, crop selection, and resource allocation well in advance.

The research paper derived from this project titled "Smart Farming through Multi Temporal Climate based AI for Agricultural Decision Support" has been accepted for oral presentation at the WAMS 2026 International Conference (IEEE), Paper ID 690. This recognition by the international research community validates the scientific quality and practical relevance of the proposed system.

Despite the promising results, several limitations of the current system warrant acknowledgment. The dataset is specific to a single geographic location in Andhra Pradesh, India, and the ACLI thresholds and weighting coefficients may require recalibration for other climatic regions. The system does not incorporate real-time data collection, direct crop type information, soil nutrient levels, or market factors all of which could enhance the completeness of decision support.

Future research directions include several promising extensions. First, the geographic coverage of the system can be expanded by training location-specific models for diverse agroclimatic zones across India, potentially leveraging transfer learning to reduce data requirements in data-scarce regions. Second, the integration of satellite-derived vegetation indices such as NDVI into the feature set could enhance the system's sensitivity to in-season crop conditions. Third, real-time deployment could be achieved through API-based automated data retrieval from NASA POWER, enabling continuous monitoring and near-real-time decision support. Fourth, the incorporation of explainable AI techniques such as SHAP (SHapley Additive exPlanations) could provide feature-level attribution for each prediction, making the system's outputs even more transparent and trustworthy for farmers and extension officers. Fifth, the system architecture could be extended to incorporate multimodal inputs combining climate data with market pricing data, crop calendars, and irrigation infrastructure maps to develop a truly comprehensive agricultural decision support platform.

In conclusion, this project demonstrates that high-quality agricultural decision support can be achieved using freely available climate data and well-established machine learning algorithms, without dependency on expensive hardware infrastructure. The proposed system is scalable, interpretable, and practically relevant, offering a viable pathway toward democratizing precision agriculture for smallholder farmers across climate-stressed agricultural regions.

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LIST OF ENCLOSURES

I. JOURNALS

1. Sharma et al. (2023) Machine learning techniques for crop yield forecasting in semi-arid (3A) zone, Rajasthan (India). Current Agriculture Research Journal.
2. Raeva et al. (2025) Yield prediction model based on multitemporal satellite data and open public data: Case study for Bulgaria. Engineering Proceedings.
3. Ali et al. (2024) Smart agriculture: Utilizing machine learning and deep learning for drought stress identification in crops. Scientific Reports.
4. Mana et al. (2024) Sustainable AI-based production agriculture: Exploring AI applications and implications in agricultural practices. Smart Agricultural Technology.
5. Islam et al. (2024) Crop yield prediction through machine learning: A path towards sustainable agriculture and climate resilience. AIMS Agriculture and Food.

II. PRESENTATIONS IN INTERNATIONAL CONFERENCES

1. Mahale et al. (2024) Crop recommendation and forecasting system using machine learning with LSTM: A novel expectation-maximization technique.
2. Gupta et al. (2025) Optimizing agricultural production using machine learning and artificial intelligence.
3. El Hachimi et al. (2023) Smart weather data management based on artificial intelligence and big data analytics for precision agriculture.
4. Jhajharia et al. (2025) A machine learning model for crop yield prediction using remote sensing data.

III. PRESENTATIONS IN NATIONAL CONFERENCES

1. Dhanaraj et al. (2025) On-device AI for climate-resilient farming with intelligent crop yield prediction using lightweight models.
2. Khaliq et al. (2025) AI-driven smart agriculture: An integrated approach for soil analysis, irrigation, and crop-fertilizer recommendations.

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ACCEPTANCE

WAMS 2026 - Notification of acceptance - Oral Presentation - Paper ID: 483 – Reg

1 message

Microsoft CMT <noreply@msr-cmt.org> Sun, Mar 1, 2026 at 11:44 AM
To: BEHARA REVATHI KUSUMA <revathikusuma65@gmail.com>

Dear Dr/Prof BEHARA REVATHI KUSUMA,
Thank you for your interest in WAMS 2026 and for submitting your paper.
Congratulations! We are pleased to inform you that, based on the reviews received, your paper has been accepted for "Oral presentation" at the symposium. Please find the details below:

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Best regards,
TPC Chair, WAMS 2026

Smart Farming through Multi Temporal Climate based AI for Agricultural Decision Support

[1]Behara Revathi Kusuma*
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
revathikusuma65@gmail.com

[2]Kondeti Akhil Lokesh
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
akhilkondeti596@gmail.com

[3]Kondaka Tarun Sai
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
kondakatarunsai@gmail.com

[4]Valurouthu Uday Chandra Naidu
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
valurouthuudaychandranaidu@gmail.com

[5]Kolli Chaithra Jyotshna
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
chaithra.kolli17@gmail.com

[6]Ramesh Yegireddi
Department of CSE
Aditya Institute of Technology and Management
Tekkali, Andhra Pradesh, India
rameshyegireddi@gmail.com

Corresponding Author: rameshyegireddi@gmail.com

***Abstract*— Agricultural productivity is largely affected by climate related factors. However, it is very hard for farmers to buy equipment for processing the soil and checking the nutrient content of the soil. In this paper, a new approach is proposed for examining agricultural land by considering environmental factors rather than chemical properties of the soil. The climate related parameters are considered as root zone soil moisture, surface soil wetness, rainfall, and temperature. The agricultural land is evaluated by calculating the agricultural condition level index (ACLI) and checking the suitability of the land as poor, moderate, and good. The decision tree algorithm is considered as a baseline, and the accuracy is enhanced by using a gradient boosting approach. The results show that the land is in a good state during the monsoon season due to rainfall, and the summer season is very stressful for the land.**

Keywords—Precision Agriculture, Climate based Decision Support, Multi temporal climate analysis, Machine Learning in Agriculture, Crop stress prediction, Irrigation Planning.

I. Introduction

Climate variability and inefficient agricultural decision-making have increasingly challenged agriculture in recent times. The climatic factors that play an important role in crop growth and stress levels include temperature, rainfall, humidity levels, and wind speed. However, the detection of climate stress levels is generally realized after crop yield reduction. Developing and rural regions generally face challenges due to the lack of timely agricultural interventions. Traditional agricultural practices generally depend on the experience of farmers or the use of sensor-based monitoring systems. The use of sensor-based systems is generally costly and challenging. Recent advances in artificial intelligence and machine learning offer new opportunities for the development of data-driven agricultural decision support systems using free climatic data. A multi temporal climate based AI framework is introduced for analyzing day wise, week wise, and month wise climatic features for agricultural condition levels and crop stress levels. The proposed AI framework uses a climate rule based index for generating pseudo labels.

II. Literature Review

Recent development in AI has influenced precision agriculture much more in crop yield prediction, stress detection, and agricultural decision support. Sharma et al. [1] (2023) proposed a machine learning model using RF, SVM, and deep learning for crop yield forecasting, achieving about 92.3% accuracy, but the study lacked decision support. Mahale et al. [2] (2024) proposed a recommendation system using RF and LSTM achieved 92% accuracy, but its reliance on short term LSTM forecasting limited its effectiveness. Raeva et al. [3] (2025) created a prediction model using Sentinel-2 imagery and ERA5 climate data achieved $R^2 = 0.78$ with regression trees but limited in estimation. Gupta et al. [4] (2025) developed a crop recommendation model using RF, SVM, and KNN improved accuracy by 12% but did not use long term climate information. Ali et al. [5] (2024) worked on drought stress detection using Gradient Boosting and LSTM & achieved 97% accuracy but was limited in yield prediction and decision support. Dhanaraj et al. [6] (2025) have worked for yield prediction using Random Forest based AI system & achieved 90.1% accuracy but relied heavily on IoT infrastructure. Mana et al. [7] (2024) A review assessed AI and data analytics applications for sustainable agricultural development. El Hachimi et al. [8] (2023) proposed an AI framework using ERA5 climate data with LSTM and XGBoost supports weather prediction and agricultural decision making. Jhajharia et al. [9] (2025) proposed a machine learning model using satellite and rainfall data predicted crop yield with SVM gaining the best performance, but it did not provide advice for farmers. Islam et al. [10] (2024) have implemented a climate-aware crop yield prediction system using machine learning achieved the best performance with XGBoost ($R^2 = 0.9745$). Khaliq et al. [11] (2025) highlighted the importance of smart farming techniques that apply artificial intelligence through the integration of information regarding the climate and soil. Their study highlighted that many methods do not incorporate multi temporal climate based decision support. Thus, this highlighted the importance of more holistic and integrated approaches to smart farming.

III. METHODOLOGY

3.1 Dataset Description

The daily climate data from the NASA POWER database (2015–2024) is used. The rainfall, temperature, and soil moisture data are used to calculate the Irrigation Index, Crop Index, and ACLI for land suitability assessment.

3.2 System Architecture Overview

The Smart Farming Decision Support System uses seasonal data on weather and soil moisture for decision making. It analyzes changes in the environment over time to assess land suitability for farming. The overall system architecture consists of the following components: (1) Acquisition of Multi Temporal climate and soil data, (2) Data Quality Assurance and Preprocessing Module to remove errors or missing values, (3) Calculating the land condition score (ACLI) which represents the overall condition of the agricultural land, and (4) Tree based Machine learning analysis and decision making to determine land suitability and classify crop growth potential into different feasibility levels.

Finally, the system outputs clear recommendations that can be used for irrigation, cropping, and other scheduling plans by farmers. This modular architecture keeps the system scalable, understandable, and suitable for real-world farming conditions; this is particularly true in those regions of the world where the chemical soil nutrient data is not available.

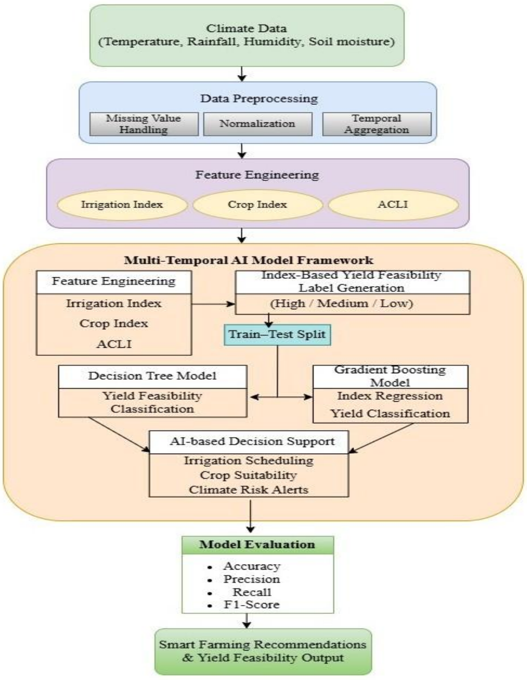

Figure 1 above shows the architecture of the proposed multi temporal climate based AI driven decision support system for smart farming

3.2.1 Acquisition of Multi-Temporal Climate and Soil Data

The first step of the methodology involves the acquisition of multi-temporal climate and soil parameters that have a direct effect on agricultural productivity. The observations are taken on a monthly basis to account for both seasonal and temporal changes.

The parameters of interest in this analysis are:

1. Root Zone Soil Wetness (RZSW)
2. Surface Soil Wetness (SSW)
3. Precipitation (P)
4. Temperature (T)
5. Temporal indicators such as month or season

These variables are essential in providing information regarding soil water availability, rainfall, and climatic stress, which are important in understanding the behavior of crop growth.

3.2.2 Data Quality Assurance and Preprocessing Module

Missing values, inconsistencies, and noise in the raw climate and soil data may impact the performance of models. Hence, a data quality assurance and preprocessing module is applied before further analysis.

Missing data is treated using statistical imputation methods like mean or median substitution. To facilitate uniform scaling of features, min-max normalization is used on all numerical attributes, given by:

1) $$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This preprocessing operation helps to ensure that all input features contribute equally to the model training process.

3.2.3 Agricultural Condition Level Index (ACLI) Computation

To represent the overall condition of agricultural land using climate-based indicators, a composite index called the Agricultural Condition Level Index (ACLI) is computed.

3.2.3.1 Feature Selection for ACLI

The ACLI calculation is dependent on climate and soil moisture variables such as Root Zone Soil Wetness, Surface Soil Wetness, Precipitation, and Temperature, which directly influence crop growth.

These factors together represent moisture availability and climatic stress.

3.2.3.2 Mathematical Formulation of ACLI

The ACLI is calculated using a weighted linear combination of the selected parameters:

2) $$ACLI = w_1 \cdot RZSW + w_2 \cdot SSW + w_3 \cdot P - w_4 \cdot T$$

where w_1, w_2, w_3,and $w_4$are weighting coefficients. Larger values of ACLI represent good land conditions, and smaller values represent environmental stress.

3.2.3.3 Normalization Prior to ACLI Calculation

All the features used in the computation of ACLI are normalized before applying the above formula to prevent any bias that may arise due to the difference in scales of the features. This step ensures consistency and comparability across different temporal observations.

3.2.3.4 ACLI based Threshold Mapping

Since actual crop yield data is not considered, the ACLI values are assigned to levels of relative land suitability based on predefined thresholds:

$$Class = \begin{cases} Poor, & ACLI < T_1 \\ Moderate, & T_1 \leq ACLI < T_2 \\ Good, & ACLI \geq T_2 \end{cases}$$

This mapping takes the continuous values of the ACLI and maps them to discrete feasibility classes, which are appropriate for supervised learning.

3.2.4 Tree based Machine learning analysis

This module represents the stage where traditional tree-based machine learning models are applied to analyze the relationship between climate variables, the Agricultural Condition Level Index (ACLI), and agricultural land suitability. The models are implemented using existing machine learning libraries, and their working principles are explained to demonstrate the inference process.

3.2.4.1 Decision Tree Classifier

A Decision Tree classifier is used as a baseline model because of its simplicity and interpretability. The model learns a set of hierarchical, rule-based decision paths that describe how various climate and soil moisture parameters influence land suitability. The optimal feature split is chosen at each internal node based on entropy and information gain, defined as:

3) $H(S) = -\sum_{i=1}^{n} p_i \log_2 (p_i)$

4) $IG(S, A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$

This process enables the model to generate transparent decision rules for classifying land conditions into poor, moderate, or good categories.

3.2.4.2 Gradient Boosting Classifier

Gradient Boosting is used to improve prediction accuracy by combining multiple decision trees. Each tree learns from the errors of the previous one to improve overall performance.The ensemble prediction is expressed as:

5) $F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$

where $F_m(x)$ represents the ensemble model at step m, $h_m(x)$ represents a weak learner or a decision tree and η represents the coefficient or parameter on which the contribution of each tree depends. It should be noted that this methodology helps to better understand complex nonlinear relationships between conditions and land suitability.

3.2.4.3 Model Comparison and Selection

Decision Tree and Gradient Boosting models were both trained using the same data set. Although the Decision Tree model is used as a baseline, the Gradient Boosting model was found to be more accurate and was selected for land suitability classification.

3.3 Training Procedure

3.3.1 Dataset Formation

After cleaning the data and calculating the ACLI value, the weather details are combined to create the final dataset with the ACLI score. Each record of data contains root zone soil wetness, surface soil wetness, precipitation, temperature, and the ACLI value. Based on these values, the land is labeled as poor, moderate, or good with respect to farming.

3.3.2 Model Training

First, a Decision Tree model is trained for a basic output. Then, a Gradient Boosting model is trained iteratively to enhance the accuracy of output. Critical parameters, such as learning rate or the number of trees to be used, can be tuned by experimentation for a better output.

3.3.3 Model Selection

Both models are compared based on common metrics of performance. The Decision Tree Model is simple and easy to interpret, but the predictions are more accurate using the Gradient Boost Model; therefore, the final model is selected as the Gradient Boost Model for predictions.

3.4 Model Evaluation Metrics

Accuracy: It is defined as an instrument's ability to measure a value accurately.

6) $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$

Precision: Based on the information its digital values convey, the precision activity demonstrates how to compare multiple measurements.

7) $Precision = \frac{TP}{TP+FP}$

Recall: The recall is the proportion of accurately anticipated positives across all class inspections.

8) $Recall = \frac{TP}{TP+FN}$

F1 score: The average of recall and precision combined.

9) $F1 = 2 \times \frac{Precision \times Recall}{Precision+Recall}$

These measures together give a comprehensive evaluation how well the classification works.

3.5 Agricultural Decision Support Output

The system provides agricultural recommendations that include irrigation, crop selection, and seasonality based on land suitability. The system is cost effective because it uses only weather and soil moisture data.

IV. Result Analysis

This section analyzes the proposed smart farming system. It evaluates the performance of Decision Tree and Gradient Boosting models in predicting climate related values and classifying land suitability.

4.1 Decision Tree Model Analysis

Firstly, the decision tree model was run to obtain a reference value to use in the prediction of climate based farming data. It was used to predict various aspects of farming and to classify the land as performing well, badly, or average based on the yield produced.

4.1.1 Regression Analysis

The Decision Tree Regressor gave consistent results in forecasting the Irrigation Index, Crop Index, and the Agricultural Climate Load Index.

Table 1: Decision Tree Regression Model Performance

Index	MAE	RMSE	R^2
Irrigation Index	0.0244	0.0350	0.9824
Crop Index	0.0290	0.0483	0.9435
ACLI	0.0250	0.0414	0.9622

The respective values of MAE and RMSE were reliably low, with coefficients of determination R^2 >0.94 in every targeted variable, therefore indicating that nonlinear relationships between climatic parameters and agricultural conditions have been accurately learnt.

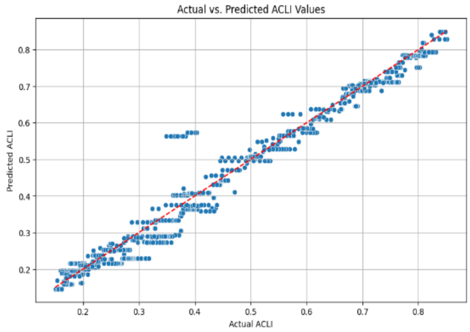

Figure 2: This graph above shows the actual vs. predicted ACLI data points using the Decision Tree Regression Model.

The Actual vs. Predicted ACLI plot shows a strong linear correlation, with most data points closely aligned along the reference line. Small deviations in the middle range might be due to transitional climatic conditions, thereby supporting the Decision Tree model.

4.1.2 Classification Analysis

For yield feasibility classification, the overall accuracy level attained by the Decision Tree model is 90.48%.

Table 2: Decision Tree Classification Model Performance

Class	Precision	Recall	F1-Score	Support
High	1.00	0.94	0.97	269
Low	0.90	0.88	0.89	269
Medium	0.83	0.90	0.86	260
Accuracy	-	-	0.9048	798
Macro Avg	0.91	0.90	0.91	798
Weighted Avg	0.91	0.90	0.91	798

From the classification graph, high precision values are obtained for the High yield class, while all the classes are balanced with regard to the recall values.

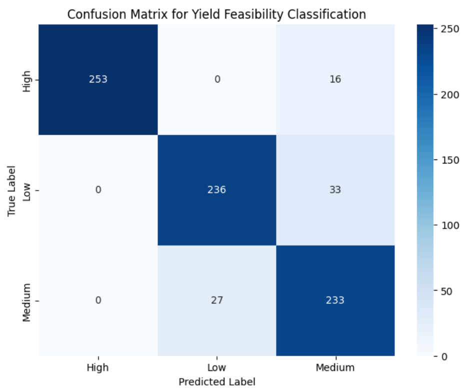

Figure 3: This confusion matrix illustrates the performance of the Decision Tree model in classifying yield feasibility classes.

As shown by the confusion matrix above, we have high diagonal values that indicate correct classification. However, some confusion between Medium and Low classes is a limitation of a single Decision Tree model.

4.2 Gradient Boosting Model Analysis

The model used to get more accurate and stable results is Gradient Boosting. It works in such a way that it fixes the earlier commitment mistakes step by step, which helps the system better understand the complex relationship between climate conditions and farming.

4.2.1 Regression Analysis

The Gradient Boosting Regressor gave consistent results in forecasting the Irrigation Index, Crop Index, and the Agricultural Climate Load Index.

Table 3: Gradient Boosting Regression Model Performance

Index	MAE	RMSE	R^2
Irrigation Index	0.0107	0.0152	0.9967
Crop Index	0.0166	0.0238	0.9863
ACLI	0.0165	0.0238	0.9875

Substantial reductions in MAE and RMSE were observed across all indices, with R^2 values reaching 0.9967 for the Irrigation Index, 0.9863 for the Crop Index, and 0.9875 for ACLI.

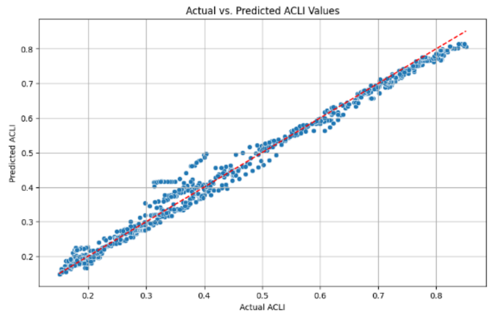

Figure 4: This graph above shows the actual vs. predicted ACLI data points using the Gradient Boosting Regression Model.

The Actual vs. Predicted ACLI plot shows that Gradient Boosting predictions are closer to the reference line, indicating higher accuracy than the Decision Tree model.

4.2.2 Classification Analysis

The accuracy achieved was 92.61% for the Gradient Boosting classifier, which shows that this model performed much better than the first simple model implemented.

Table 4: Gradient Boosting Classification Model Performance

Class	Precision	Recall	F1-Score	Support
High	1.00	0.96	0.98	269
Low	0.94	0.87	0.91	269
Medium	0.85	0.95	0.89	260
Accuracy	-	-	0.9261	798
Macro Avg	0.93	0.93	0.93	798
Weighted Avg	0.93	0.93	0.93	798

Precision, recall, and F1-score values also clearly indicate that the model performed fairly for all classes poor, moderate, and good of lands.

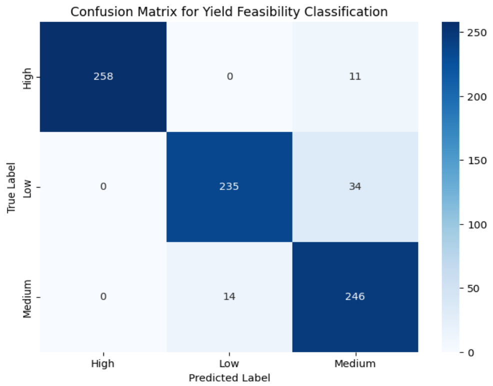

Figure 5: This confusion matrix illustrates the performance of the Gradient Boosting model in classifying yield feasibility classes.

From the above confusion matrix, it may be queried that there is a decrease in misclassification of Medium and Low yield classes; meanwhile, zero confusion is maintained for High and Low classes. Similarly, there is an improvement in recall value for the Medium class. This reflects that generalization of classes under moderate climatic conditions is difficult.

V. Discussion

The proposed multi temporal climate based AI framework is effective in supporting decision making in the agricultural sector by learning long term climatic patterns and incorporating seasonal patterns. Unlike other decision-making techniques and models, the proposed model uses multi-temporal climate data for decision-making, ensuring more reliable and stable decision-making. A comparative analysis between the Decision Tree and Gradient Boosting techniques is presented in the following figure, where the Decision Tree model is effective in decision-making by generating decision rules. However, the model is unable to handle complex nonlinear relationships. On the other hand, the Gradient Boosting model is effective in decision-making by generating decision rules, and its performance is high in all the evaluation parameters, ensuring improved prediction accuracy. Compared with other research studies, the proposed model is effective in decision-making in the agricultural sector by providing decision-making insights for farmers based on historical climate data. Additionally, the model is effective in avoiding expensive IoT devices for collecting climate data, as the model uses free climate data for decision-making. Therefore, the proposed model is effective in decision-making in the agricultural sector by incorporating multi-temporal ensemble learning, and the model bridges the gap between climate analytics and decision-making in the agricultural sector.

VI. Conclusion

In this regard, this specific paper outlines a smart farming strategy that assists farmers in making good and adaptable decisions through the use of an artificial intelligence system based on long-term climate information, as opposed to specific data or information based on weather changes, which are commonly derived from pre-defined data as well as from short-term weather information updates. When the different models were compared based on the different tests carried out, Gradient Boosting performed much better compared to the Decision Tree Model. It showed higher accuracy and consistency in its results compared to the Decision Tree Model. Although the latter is an easy model to understand and can be easily explained with a series of decisions made based on the different features involved in the data, the higher complexity shown by the different relationships in the climate makes the model less reliable for long-term decisions related to farming. Unlike most of the existing approaches, which often look at only yield prediction or short-term forecasts, this work takes an extra step towards practical decision support for agriculture. This developed framework makes use of both long-term climatic data and ensemble learning methods to provide definite and scalable farming practices. It is simultaneously not depending highly on sensor information, making it more appropriate for farming regions that are poor in resources. Overall, this paper demonstrated that using long-term climatic information in artificial intelligence-based models makes a smart farming system more effective in general. The framework that was developed links climatic analysis to various farming applications in order to support sustainable and climatically resilient agriculture in general. Future research could focus on adding more data in terms of soil, satellite images, and market data. The model structure will be improved to employ real-time learning and tested over a wider area.

VII. References

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